MESWebApi Method List

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MESWebApi Method List

Assembly

1. GetAssemblyById

Api/Method:

- Assembly/GetAssemblyById

Input Paramters:

- FromBody]GetAssemblyById
    - assemId – assembly_id
    - langId – “0” is not localized or English
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- Assembly_ID
- AssemblyName
- Number
- Revision
- Descr
- Description
- Phantom
- BOM_ID
- Material
- EffectiveFrom
- EffectiveTo
- Active
- Customer_ID
- CustomerName
- DivisionName
- Panel_ID
- PanelName
- Family_ID
- FamilyName
- BarcodeMask_ID
- BarcodeText
- UseMultiPartBarCode
- UserID_ID
- LastUpdated
- CheckDate

Preconditions:

Note:

2. GetAssemblyProgressionByAssemblyRouteStep

Api/Method:

- Assembly/GetAssemblyProgressionByAssemblyRouteStep

Input Paramters:

- [FromBody]GetAssemblyData
  - assemId – required
  - routeStepId – required
  - custId – “0” if not specific
  - langId – “0” is not localized or English
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - FromAssembly_ID
  - FromNumber
  - FromRevision
  - FromVersion
  - FromAssembly
  - FromCustomer_ID
  - FromCustomer
  - FromBOM_ID
  - FromBOM
  - ToAssembly_ID
  - ToNumber
  - ToRevision
  - ToVersion
  - ToAssembly
  - ToCustomer_ID
  - ToCustomer
  - ToBOM_ID
  - ToBOM
  - RouteStep_ID
  - FactoryMARoute_ID
  - FactoryText

- MAText
- RouteText
- StepText
- DescrText
- UserID_ID
- LastUpdated
- CheckDate

Preconditions:

Note:

3. GetAssemblyProgressionRecursive

Api/Method:

- Assembly/GetAssemblyProgressionRecursive

Input Paramters:

- [FromBody]GetAssemblyData
    - assemId
    - custId – required and validated
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
    - FromAssembly_ID
    - FromNumber
    - FromRevision
    - FromVersion
    - ToAssembly_ID
    - ToNumber
    - ToRevision
    - ToVersion
    - AssemblyLevel

Preconditions:

Note:

- If assemId is invalid, raise error with proper message
- If assemId doesn't belong to custId, raise error with proper message
- If assmId is not active, raise error with proper message

- Only active assemblies (both from and to) belong to the same custId will be return
- If no assembly progression is found, return 0 row.
- **First progression from input assembly_id is at level 0; the assembly level can increment, or decrement based on direction of assembly progression**

4. GetParentAssemblyRecursive

Api/Method:

- Assembly/GetParentAssemblyRecursive

Input Paramters:

- [FromBody]GetAssemblyData
  - assemId
  - custId – required and validated
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - Assembly_ID
  - Number
  - Revision
  - Version
  - RequiredQuantity
  - ParentAssembly_ID
  - ParentNumber
  - ParentRevision
  - ParentVersion
  - AssemblyLevel

Preconditions:

Note:

- If assemId is invalid, raise error with proper message
- If assemId doesn't belong to custId, raise error with proper message
- If assmId is not active, raise error with proper message
- Only active parent assembly belong to the same custId will be return
- If no parent is found, return 0 row.
- **Input assembly_id is at level 0; this level must be increased by one for each next parent**

5. ListAssembly

Api/Method:

- Assembly/ListAssembly

Input Paramters:

- [FromBody]GetAssemblyData
  - custId – required and validated
  - active – "1" is active
  - partialKey – "" is not specific
  - langId – "0" is not localized or English
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - Assembly_ID
  - AssemblyName
  - Number
  - Revision
  - Version
  - Descr
  - Description
  - Phantom
  - BOM_ID
  - Material
  - EffectiveFrom
  - EffectiveTo
  - Active
  - Customer_ID
  - CustomerText
  - Panel_ID
  - PanelName
  - Family_ID
  - FamilyName
  - BarcodeMask_ID
  - BarcodeMaskText
  - UseMultiPartBarCode
  - UserID_ID
  - LastUpdated
  - CheckDate

Preconditions:

Note:

6. ListAssemblyBarcodeMask

Api/Method:

- Assembly/ListAssemblyBarcodeMask

Input Paramters:

- [FromBody]ListAssemblyBarcodeMask
  - assemId
  - partialKey – "" is not specific
  - langId – “0” is not localized or English
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - BarcodeMask_ID
  - BarcodeMaskText
  - Mask

Preconditions:

Note:

7. ListAssemblyDataByCustomer

Api/Method:

- Assembly/ListAssemblyDataByCustomer

Input Paramters:

- [FromBody]ListAssemblyDataByCustomer
  - customerId – string[], e.g., [2,3]
  - assemblyStatus – e.g., “Active”
  - updatedAfter – e.g., “2019-07-01”
  - langId – “0” is not localized or English
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - CustomerId
  - CustomerName
  - DivisionName

- AssemblyNumber
- AssemblyRevision
- ParentAssemblyNumber
- ParentAssemblyRevision
- Quantity
- AssemblyFamily
- AssemblyDescription
- AssemblyStatus
- UpdatedDate

Preconditions:

Note:

8. ListAssemblyMaterialByAssemblyAndDeviation

Api/Method:

- Assembly/ListAssemblyMaterialByAssemblyAndDeviation

Input Paramters:

- [FromBody]
    - AssemblyNumber
    - AssemblyRevision
    - AssemblyVersion
    - DeviationNumber – optional, if blank, "none" will be used
    - langId – optional, if blank 0 will be used
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
    - AssemblyMaterial_ID
    - Assembly_ID
    - Number
    - Revision
    - Version
    - BuildStep
    - RequiredQuantity
    - PackoutQuantity
    - ScanPart
    - ScanRev
    - Dynamic
    - Serializd
    - Verifiable
    - Deviation_ID
    - DeviationNumber

- LinkMaterial_ID
    - LinkObject_ID
    - LinkObject
    - LinkObjectRevision
    - UserID_ID
    - CheckDate
    - Alternates
    - LinkMask
    - RouteStep_ID
    - FactoryText
    - MAText
    - RouteText
    - StepText

Preconditions:

Note:

9. ListAssemblyPropertyByAssembly

Api/Method:

- Assembly/ListAssemblyPropertyByAssembly

Input Paramters:

- [FromBody]ListAssemblyPropertyByAssembly
    - usrId – required and validated
    - number – assembly number
    - revision – assembly revision
    - langId – “0” is not localized or English
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
    - Assembly_ID
    - Number
    - Revision
    - Version
    - Assembly
    - Descr
    - Description
    - Phantom
    - BOM_ID
    - EffectiveFrom

- EffectiveTo
- Active
- Customer_ID
- Customer
- Division
- Panel_ID
- PanelType
- Family_ID
- Family
- BarcodeMask_ID
- Barcode
- UseMultiPartBarCode
- Property_ID
- Property
- PropertyValue

Preconditions:

Note:

10. ListAssemblyPropertyByAssemblyId

Api/Method:

- Assembly/ListAssemblyPropertyByAssemblyId

Input Paramters:

- [FromBody]ListAssemblyPropertyByAssemblyId
  - usrId – required and validated
  - assemId
  - partialKey – required, it is used for SQL like. If you didn't pass in exact property name, please add "%" at the end of partialKey
  - langId – "0" is not localized or English
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - Assembly_ID
  - Number
  - Revision
  - Version
  - Assembly
  - Descr
  - Description
  - Phantom

- BOM_ID
- EffectiveFrom
- EffectiveTo
- Active
- Customer_ID
- Customer
- Division
- Panel_ID
- PanelType
- Family_ID
- Family
- BarcodeMask_ID
- Barcode
- UseMultiPartBarCode
- Property_ID
- Property
- PropertyValue

Preconditions:

Note:

11. ListAssemblyRouteByAssembly

Api/Method:

- Assembly/ListAssemblyRouteByAssembly

Input Paramters:

- [FromBody]ListAssemblyRouteByAssembly
  - assemId
  - fmaRoute
  - langId – "0" is not localized or English
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - Assembly_ID
  - AssemblyName
  - Number
  - Revision
  - Version
  - Customer_ID

- FactoryMARoute_ID
- FactoryName
- ManufacturingAreaName
- RouteName
- StartDate
- FinishDate
- EstimatedUnits
- EstimatedTime
- UserID_ID
- LastUpdated
- FactoryMA_ID
- CheckDate

Preconditions:

Note:

12. ListByCustomer

Api/Method:

- Assembly/ListAssemblyForMIITool

Input Paramters:

- [FromBody]
  - usrId
  - customerId – required and validated
  - partialKey – “” is not specific
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - Assembly_ID
  - Number
  - Revision
  - Version
  - Assembly
  - Phantom
  - BOM_ID
  - SAP_BOM
  - EffectiveFrom
  - EffectiveTo
  - Active

- Customer_ID
- Division
- Panel_ID
- Family_ID
- Family
- BarcodeMask_ID
- UseMultiPartBarCode

Preconditions:

Note:

Batch

13. AddBatchPulling

Api/Method:

- Batch/AddBatchPulling

Input Paramters:

- [FromBody]AddBatchPulling
  - usrId – required and validated
  - fromBatchId – required. BatchID_ID where initial batch
  - toBatchId – required. BatchID_ID of end batch
  - routeStepId
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

Note:

- Batch Pulling configuration is stored in dbo.WP_BatchPulling table, which is outside of standard MES.exe
- fromBatchId, toBatchId and routeStepId constitute a unique key

14. DeleteBatchPulling

Api/Method:

- Batch/DeleteBatchPulling

Input Paramters:

- [FromBody]DeleteBatchPulling
    - usrId – required and validated
    - batchPullingId – required. BatchPulling_ID in dbo.WP_BatchPulling table
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

Note:

- only unused configuration (no record found in dbo.WP_WipBatchPullingHistory) can be deleted

15. GetBatchById

Api/Method:

- Batch/GetBatchById

Input Paramters:

- [FromBody]GetBatchById
    - batchId – batchID_ID
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- BatchID_ID
- BatchID
- Description
- ExpStartDateTime
- ExpEndDateTime
- ExpQty
- ActStartDateTime
- ActEndDateTime
- ActQty
- Customer_ID
- EnforceQuantity

- EnforceDates
- EnforceQuantityByAssembly
- AutoAssociateProgAsm
- UserID_ID
- LastUpdated
- CheckDate

Preconditions:

Note:

16. GetBatchByNumber

Api/Method:

- Batch/GetBatchByNumber

Input Paramters:

- [FromBody]GetBatchByNumber
  - batch
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- BatchID_ID
- BatchID
- Description
- ExpStartDateTime
- ExpEndDateTime
- ExpQty
- ActStartDateTime
- ActEndDateTime
- ActQty
- Customer_ID
- EnforceQuantity
- EnforceDates
- EnforceQuantityByAssembly
- AutoAssociateProgAsm
- UserID_ID
- LastUpdated
- CheckDate

Preconditions:

Note:

17. ListBatchAssemblyQty

Api/Method:
- Batch/ListBatchAssemblyQty

Input Paramters:
- [FromBody]ListBatchAssemblyQty
  - batchId – batchID_ID
  - assemId – “0” if to retrieve qty for all assemblies associated with the batch
  - langId – “0” is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:
- List Of
  - BatchID_ID
  - BatchID
  - Assembly
  - Number
  - Revision
  - Version
  - AssemblyText
  - Assembly_ID
  - EnforceQuantity
  - EnforceQuantityByAssembly
  - BatchExpQty
  - BatchActQty
  - ExpQty
  - ActQty
  - UserID_ID
  - LastUpdated

Preconditions:

Note:

18. ListBatchByAssembly

Api/Method:

- Batch/ListBatchByAssembly

Input Paramters:

- [FromBody]ListBatchByAssembly
    - custId -- required
    - assemId -- required
    - partialKey -- required
    - langId – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
    - BatchID_ID
    - BatchID
    - Descr
    - ExpStartDateTime
    - ExpEndDateTime
    - ExpQty
    - EnforceDates
    - ActStartDateTime
    - ActEndDateTime
    - ActQty
    - UserID_ID
    - BatchCustomer_ID
    - EnforceQuantity
    - EnforceQuantityByAssembly
    - AutoAssociateProgAsm
    - BatchCustomerText
    - Assembly_ID
    - Assembly
    - Number
    - Revision
    - Version
    - AssemblyDescr
    - Phantom
    - BOM
    - EffectiveFrom
    - EffectiveTo
    - Active
    - AssemblyCustomer_ID
    - Panel
    - Family
    - BarcodeMask_ID
    - UserMultiPartBarcode

- AssemblyCustomerText

Preconditions:

Note:

- This function will return up-to 400 rows of data
- Please set the partial key as precise as possible

19. ListBatchByAssemblyAndExpEndTime

Api/Method:

- Batch/ListBatchByAssemblyAndExpEndTime

Input Paramters:

- [FromBody]ListBatchByAssemblyAndExpEndTime
  - custId -- required
  - assemId -- required
  - expEndTime -- required
  - langId – “0” is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
  - BatchID_ID
  - BatchID
  - Descr
  - ExpStartDateTime
  - ExpEndDateTime
  - ExpQty
  - EnforceDates
  - ActStartDateTime
  - ActEndDateTime
  - ActQty
  - UserID_ID
  - BatchCustomer_ID
  - EnforceQuantity
  - EnforceQuantityByAssembly
  - AutoAssociateProgAsm
  - BatchCustomerText
  - Assembly_ID
  - Assembly

- Number
- Revision
- Version
- AssemblyDescr
- Phantom
- BOM
- EffectiveFrom
- EffectiveTo
- Active
- AssemblyCustomer_ID
- Panel
- Family
- BarcodeMask_ID
- UserMultiPartBarcode
- AssemblyCustomerText

Preconditions:

Note:

20. ListBatchByDate

Api/Method:

- Batch/ListBatchByDate

Input Paramters:

- [FromBody]
    - userId – required, must be valid user ID greater than 1
    - StartDate – required
    - EndDate – required
    - customerId – required
    - Number – optional, can be blank string
    - Revision – optional, can be blank string
    - Version – optional, can be blank string
    - MaxCount – 0 for no limit
    - Language_ID – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
    - BatchID_ID
    - BatchID
    - Descr

- ExpStartDateTime
- ExpEndDateTime
- ExpQty
- ActStartDateTime
- ActEndDateTime
- ActQty
- Customer_ID
- Assembly_ID

Preconditions:

Note:

21. ListBatchCountsByRouteStep

Api/Method:

- Batch/ListBatchCountsByRouteStep

Input Paramters:

- [FromBody]
  - CustomerID – required
  - StartDate – required
  - EndDate – required
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
  - BatchID
  - Assembly
  - RouteStep
  - Bay
  - SAP_BOM
  - StepOrder
  - ActualQty
  - FirstScanDTS
  - LastScanDTS

Preconditions:

Note:

Bay = Manufacturing Area in MES

Due to the complexity of the query, this API will only run for a maximum 24 hour period on production servers.
If a greater than 24 hour period is selected, the result will default to an end date 24 hours from the start date.
Reporting servers etc. are not affected.

- Please work with your application admin to make sure sure the web.config of api have the following setting:
  - configuration setting, sqlReportingTimeout, is used to alter SQL command timeout for GetReport method
    - Timeout in seconds.
    - E.g. 300. Please note that 0 will not be accepted. In the case 0 is set, the api will assume default behavior of 30 secs timeout

22. ListBatchForPullingByAssemblyId

Api/Method:

- Batch/ListBatchForPullingByAssemblyId

Input Paramters:

- [FromBody]ListBatchForPullingByAssemblyId
  - custId – required
  - assemId – optional
  - routeStepId – optional
  - isActive – optinal; if we only list batches of active BatchPulling configurations
  - maxCount -- optional
  - langId – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
  - Assembly_ID
  - Number
  - Revision
  - Version
  - Customer_ID
  - CustomerText
  - DivisionText
  - Family_ID
  - FamilyText
  - BarcodeMask_ID
  - UseMultiPartBarCode

- CustomerNumber
    - CustomerRevision
    - BuildStatus_ID
    - BuildStatus
    - BatchPulling_ID
    - ToBatchID_ID
    - ToBatchID
    - ToBatchExpQty
    - ToBatchActQty
    - FromAssembly_ID
    - FromNumber
    - FromRevision
    - FromVersion
    - FromBatchID_ID
    - FromBatchID
    - RouteStep_ID
    - UserID_ID
    - LastUpdated
    - CheckDate

Preconditions:

Note:

- Pass in valid assembly_id and routestep_id for more efficient search

23. ListBatchPulling

Api/Method:

- Batch/ListBatchPulling

Input Paramters:

- [FromBody]ListBatchPulling
    - fromBatch – optional, initial BatchID
    - toBatch – optional, end BatchID
    - status – optional. Active, Hold, Canceled or Fulfilled; default is Active
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
    - BatchPulling_ID
    - FromBatch
    - FromBatchExpQty
    - FromBatchActQty

- ToBatch
- ToBatchExpQty
- ToBatchActQty
- MAText
- RouteText
- StepText
- RouteStep_ID
- DescrText
- Active
- Status

Preconditions:

Note:

24. ListWipBatchPullingHistory

Api/Method:

- Batch/ListWipBatchPullingHistory

Input Paramters:

- [FromBody]ListWipBatchPullingHistory
  - wipId
  - serial – wipId or serial must be provided
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
  - Wip_ID
  - SerialNumber
  - WipBatchPullingHistory_ID
  - BatchPulling_ID
  - FromBatchID_ID
  - FromBatchID
  - FromAssembly_ID
  - FromAssembly = Number/Revision/Version
  - ToBatchID_ID
  - ToBatchID
  - ToAssembly_ID
  - ToAssembly = Number/Revision/Version
  - RouteStep_ID
  - RouteStepText = MAText/RouteText/StepText
  - UserID_ID

- LastUpdated
- CheckDate

Preconditions:

Note:

25. UpdateBatchPullingStatus

Api/Method:

- Batch/UpdateBatchPullingStatus

Input Paramters:

- [FromBody]UpdateBatchPullingStatus
  - usrId – required and validated
  - batchPullingId – required
  - routeStepId – required; validated
  - status – Active, Hold or Canceled
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

Note:

- api to inactivate (hold or cancel) or activate (active) a batch pulling configuration
- to update RouteStep_ID for a particular BatchPulling, provide new routeStepId
- to update BatchPulling status only, provide current routeStepId for validation

26. WipBatchPulling

Api/Method:

- Batch/WipBatchPulling

Input Paramters:

- [FromBody]WipBatchPulling
  - custId – required and validated
  - usrId – required and validated
  - wipId – required

- batchPullingId – required; BatchPulling_ID selected
    - toBatchId – requires; end BatchID_ID
    - toAssemblyId – required; end Assembly_ID
    - routeStepId
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

Note:

- Assign Board to ToBatch
- ToBatch ActQty incremented by 1
- FromBatch's ActQty remains the same to avoid aging inventory concern -- boards left in FromBatch for longer than intended
- When ToBatch's ActQty = ExpQty, set the associated BatchPulling configuration with Status == Fulfilled and Active == 0
- Log Wip batch change history to dbo.WP_WipBatchHistory table
- Process verification, Wip Move, and Assembly Progression remain unchanged, and not included in this stored procedure.

27. WipBatchPullingReversal

Api/Method:

- Batch/WipBatchPullingReversal

Input Paramters:

- [FromBody]WipBatchPullingReversal
    - custId – required and validated
    - usrId – required and validated
    - serial – required; Wip serial number
    - routeStepId
    - assemId – if 0 reverse back to birthing BatchID_ID and Assembly_ID; otherwise, reverse back to the associated BatchID_ID associated with assemId
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

Note:

- Assign Board to its original batch and assembly
- Current batch ActQty decremented by 1
- Orignal ActQty remains the same to avoid aging inventory concern – it was not decremented when batch pulling occurred
- Change batchpulling configuration to Active after batchpulling reversal if batch pulling is at “Fulfilled” status
- Log Wip batch change history to dbo.WP_WipBatchHistory table

Bom

28. GetBOMMaterialsByBOM

Api/Method:

- Bom/GetBOMMaterialsByBOM

Input Paramters

- [FromBody]
  - BOM_ID
  - Language_ID – “0” is not localized or English
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
  - BOM_ID
  - BOMMaterial_ID
  - BOMMaterial
  - Material_ID
  - Material
  - Description
  - Qty
  - EffectiveFrom
  - EffectiveTo
  - BOMLevel
  - Assembly
  - BOMSortOrder
  - UserID_ID
  - Checkdate

Preconditions:

Note:

29. ListBOM

Api/Method:

- Bom/ListBOM

Input Paramters

- [FromBody]
    - Material
    - MaxCount
    - Language_ID
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
    - BOM_ID
    - BOMMaterial_ID
    - Material
    - Descr
    - Validated
    - UserID_ID
    - CheckDate

Preconditions:

Note:

Container

30. GetBoxContents

Api/Method:

- Container/GetBoxContents

Input Paramters

- [FromBody]GetBoxContents

- custId – required and validated
- box– required
- langId – “0” is not localized or English – optional

- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
  - Container _ID
  - ContainerNumber CustomerText
  - WIP_ID
  - SerialNumber
  - Assembly_
  - Number
  - Revision
  - Version
  - PackDate

Preconditions:

Note:

31. GetBoxContentsWithProperties

Api/Method:

- Container/GetBoxContentsWithProperties

Input Paramters

- [FromBody]
  - JsonString
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

Preconditions:

Note:

32. GetBoxDataByBoard

Api/Method:

- Container/ GetBoxDataByBoard

<u>Input Paramters</u>

- [FromBody] GetBoxDataByBoard
    - custId – required and validated
    - serial– required
    - langId – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

<u>Expected Result</u>:

- Container_ID
- Customer_ID
- CustomerText
- DivisionText
- FamilyText
- ContainerNumber
- ContainerSubType_ID
- ContainerSubTypeText
- ContainerType_ID
- ContainerTypeText
- Assembly_ID
- AssemblyNumber
- AssemblyRevision
- BuildStatus_ID
- BuildStatusText
- BoardCount
- ContainerCapacity
- ContentRule_ID
- ContentRuleText
- ContainerStatus_ID
- ContainerStatusText
- CreatedDate
- UserID_ID
- LastUpdated
- CheckDate

<u>Preconditions:</u>

Note:

Customer

33. ListCustomer

Api/Method:

- Customer/ListCustomer

Input Paramters:

- [FromBody]ListCustomerConfig
  - partialKey– required
  - active= "1" if active only
  - langId – “0” is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
  - Customer_ID
  - Customer
  - CustomerName
  - Division
  - DivisionName
  - CustomerDivisionName
  - SAPIdentifier
  - ForceValidGRN
  - ProhibitLastPart
  - RequireQuickscan
  - UserID_ID
  - LastUpdated
  - CheckDate
  - Active

Preconditions:

Note:

34. ListCustomerByFilter

Api/Method:

- Customer/ListCustomerByFilter

Input Paramters:

- [FromBody]ListCustomerByFilter
  - customerStatus– e.g., “Active”

- updatedAfter – e.g., "2019-07-01"
- langId – "0" is not localized or English – optional

- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
  - Customer_ID
  - CustomerName
  - CustomerStatus
  - DivisionName
  - LastUpdated

Preconditions:

Note:

35. ListCustomerConfig

Api/Method:

- Customer/ListCustomerConfig

Input Paramters:

- [FromBody]ListCustomerConfig
  - custId – required and validated
  - usrId – required and validated
  - partialKey– required
  - langId – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
  - Customer_ID
  - Customer
  - CustomerText
  - Division
  - DivisionText
  - KeyDescription
  - KeyValue
  - UserID_ID

Preconditions:

Note:

DB

36. RestoreSNHistoryWByWipID

Api/Method:

- DB/ RestoreSNHistoryWByWipID

Input Paramters:

- [[FromBody]RestoreSNHistoryWByWipID
    - wipSNs – Wip serials sparated by “|”
    - usrId – UserID_ID
    - mode – optional; default is 0
    - custId -- optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

Note:

Mode:

- 0 = Quiet, ReturnCode only
- 1 = Interactive (Upgrade Manager),
- 2 = Insert into a WP_WipsNoPurge Table with 1,1,1, also mode 2 is executed by SQL Job and SP "AUS_RestorePendingSNHistoryWByWip"

Equipment

37. GetEquipmentById

Api/Method:

- Equipment/GetEquipmentById

Input Paramters:

- [FromBody]GetEquipmentById
    - usrId

- equipId – equipment_id
- langId – “0” is not localized or English – optional

- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- Equipment_ID
- EquipmentMaster_ID
- Make
- Vendor
- Model
- CommonName
- SeqNumber
- AssetTag
- ManSerialNumber
- Available
- EquipmentSetup_ID
- TableA
- TableB
- PPM
- DateOfManufacture
- RunTimeHours
- SoftwareVersion
- Condition_ID
- ConditionText
- Status_ID
- StatusText
- FeederTracking
- FeederTypeValidation
- UserID_ID
- LastUpdated
- CheckDate

Preconditions:

Note:

38. GetEquipmentByName

Api/Method:

- Equipment/GetEquipmentByName

Input Paramters:

- [FromBody]GetEquipmentByName
  - commonName – equipment common name
  - langId – “0” is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- Equipment_ID
- EquipmentMaster_ID
- Make
- Vendor
- Model
- CommonName
- SeqNumber
- AssetTag
- ManSerialNumber
- Available
- EquipmentSetup_ID
- TableA
- TableB
- PPM
- DateOfManufacture
- RunTimeHours
- SoftwareVersion
- Condition_ID
- ConditionText
- Status_ID
- StatusText
- FeederTracking
- FeederTypeValidation
- UserID_ID
- LastUpdated
- CheckDate

Preconditions:

Note:

39. GetEquipmentSetupGRNQty

Api/Method:

- Equipment/GetEquipmentSetupGRNQty

Input Paramters:

- [FromBody]GetEquipmentSetupGRNQty
    - usrId – required and validated
    - routeStepId – RouteStep_ID
    - assemId – Assembly_ID for the serial number
    - equipId – Equipment_ID
    - equipSetupId – EquipmentSetup_ID
    - grn– "" if not specific
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of (GRN, Material, Qty)

Preconditions:

- Equipmentsetup must be completed in MES

Note:

- If grn = "", List of (GRN, Material, Qty). Else if grn is specified, should have only 1 row

40. GetFeederTrayTrackByName

Api/Method:

- Equipment/GetFeederTrayTrackByName

Input Paramters:

- [FromBody]GetFeederTrayTrackByName
    - feeder
    - tray
    - track
    - langId – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- FeederTrayTrack_ID
- Feeder
- Tray
- Track
- UserID_ID
- LastUpdated
- CheckDate

Note:

41. GetTransportByName

Api/Method:

- Equipment/GetTransportByName

Input Paramters:

- [FromBody]GetTransportByName
  - tranport
  - langId – “0” is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- Transport_ID
- EquipmentMaster_ID
- Make
- Vendor
- Model
- Transport
- AssetTag
- ManSerialNumber
- Status_ID
- StatusText
- Condition_ID
- ConditionText
- Comments
- UserID_ID
- LastUpdated
- CheckDate

Note:

42. ListEquipmentByRouteStep

Api/Method:

- Equipment/ListEquipmentByRouteStep

Input Paramters:

- [FromBody]ListEquipmentByRouteStep
    - routeStepId – required and validated
    - langId – “0” is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
    - RouteStep_ID
    - FactoryName
    - ManufacturingAreaName
    - RouteName
    - StepName
    - RouteStepDescription = p.Translation
    - Occurrence
    - Equipment_ID
    - Vendor
    - Model
    - CommonName
    - EquipmentRow
    - EquipmentColumn
    - ProcessingTime
    - UserID_ID
    - LastUpdated
    - CheckDate

Note:

EquipmentSetup

43. ClearSetupTable

Api/Method:

- EquipmentSetup/ClearSetupTable

Input Paramters:

- [FromBody]ClearSetupTable
    - usrId – required and validated
    - setupId – equipment setup Id
    - equipId
    - clearMachine
    - table
    - removeTransport

- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:
- success/error

Preconditions:

Note:

44. DeleteEntireSetup

Api/Method:
- EquipmentSetup/ClearSetupTable

Input Paramters:
- [FromBody]
  - EquipmentSetup_ID
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:
- success/error

Preconditions:

Note:

45. EquipmentComponentAddPartByLane

Api/Method:
- EquipmentSetup/EquipmentComponentAddPartByLane

Input Paramters:
- [FromBody]EquipmentComponentAddPartByLane
  - usrId – required and validated
  - setupId – equipmentsetup_ID
  - equipId
  - feederTrayTrackId,
  - grnId
  - qty
  - tranportId
  - setupFeederType

- loadedFeederType
- table
- installed
- lane

- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

Note:

46. EquipmentComponentRemovePart

Api/Method:

- EquipmentSetup/EquipmentComponentRemovePart

Input Paramters:

- [FromBody]EquipmentComponentRemovePart
  - usrId – required and validated
  - equipId
  - feederTrayTrackId,
  - grnId
  - table
  - removeTransport
  - reelIsEmpty
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

Note:

47. GetActiveEquipmentSetupByAssemblyId

Api/Method:

- EquipmentSetup/GetActiveEquipmentSetupByAssemblyId

Input Paramters:

- [FromBody]
    - Assembly_ID
    - MaxCount – optional, 0 returns all records
    - Language_ID - “0” is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
    - EquipmentSetup_ID
    - RouteStep_ID
    - Equipment_ID
    - Assembly_ID
    - Number
    - Revision
    - Version
    - AssemblyText
    - FactoryText
    - MAText
    - RouteText
    - StepText
    - CommonName
    - Vendor
    - Model
    - BOM_ID
    - SetupNumber
    - SetupVersion
    - TotalComponents
    - TotalPins
    - Export
    - Active
    - RFDisplay
    - MachineName
    - ProgramName
    - LoadDateTime
    - Assembly
    - CommonEquipmentSetup_ID
    - CommonSetupName
    - UserID_ID
    - CheckDate

Preconditions:

Note:

48. GetEquipmentSetupByEquipmentAssembly

Api/Method:

- EquipmentSetup/GetEquipmentSetupByEquipmentAssembly

Input Paramters:

- [FromBody]
  - Equipment_ID
  - Assembly_ID
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - EquipmentSetup_ID
  - RFDisplay
  - SetupNumber
  - SetupVersion
  - MaxVersion
  - MachineName
  - ProgramName
  - ForceValidGRN
  - ProhibitLastPart

Preconditions:

Note:

49. GetEquipmentSetupById

Api/Method:

- EquipmentSetup/GetEquipmentSetupById

Input Paramters:

- [FromBody]
  - Equipment_ID
  - Language_ID – "0" is not localized or English – optional
- sqlServer = "" – optional

- dataBase = "" – optional

Expected Result:

- List of
  - EquipmentSetup_ID
  - RouteStep_ID
  - Equipment_ID
  - Assembly_ID
  - Number
  - Revision
  - Version
  - AssemblyText
  - FactoryText
  - MAText
  - RouteText
  - StepText
  - CommonName
  - Vendor
  - Model
  - BOM_ID
  - SetupNumber
  - SetupVersion
  - TotalComponents
  - TotalPins
  - Export
  - Active
  - RFDisplay
  - MachineName
  - ProgramName
  - LoadDateTime
  - Assembly
  - CommonEquipmentSetup_ID
  - CommonSetupName
  - UserID_ID
  - CheckDate

Preconditions:

Note:

50. GetNewSetups

Api/Method:

- EquipmentSetup/GetNewSetups

Input Paramters:

- [FromBody]
  - MachineName
  - MaxCount – optional, 0 is all records
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - EquipmentSetup_ID
  - MachineName
  - ProgramName
  - LoadDateTime
  - Assembly
  - TotalComponents
  - UserID_ID
  - CheckDate

Preconditions:

Note:

51. InsertEquipmentSetup

Api/Method:

- EquipmentSetup/InsertEquipmentSetup

Input Paramters:

- [FromBody]
  - EquipmentSetup_ID – required, must be ID already in CT_Setup table but not in CT_EquipmentSetup (see EquipmentSetup/InsertSetup API method)
  - RouteStep_ID
  - Equipment_ID
  - Assembly_ID
  - SetupNumber
  - TotalComponents
  - TotalPins
  - Export
  - Active
  - RFDisplay
  - CommonEquipmentSetup_ID

- UserID_ID – required, must be valid user
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:
Success: Will return 1
Failure: Will return error message

Preconditions:

Note:

52. InsertEquipmentSetupFeeder

Api/Method:

- EquipmentSetup/InsertEquipmentSetupFeeder

Input Paramters:

- [FromBody]
  - EquipmentSetup_ID– required, must be valid equipment setup ID
  - Feeder
  - Tray
  - Track
  - FeederType
  - Media
  - Material_ID
  - MaterialDescr
  - CRDCount
  - OutOfStock
  - Deviation
  - Change
  - UserID_ID – required, must be valid user
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:
Success: Will return new Feeder_ID
Failure: Will return error message

Preconditions:

Note:

53. InsertFeederCRD

Api/Method:

- EquipmentSetup/InsertFeederCRD

Input Paramters:

- [FromBody]
  - Feeder_ID– required, must be valid feeder ID
  - CRD
  - UserID_ID – required, must be valid user
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:
Success: Will return 1
Failure: Will return error message

Preconditions:

Note:

54. InsertFeederCRDOffset

Api/Method:

- EquipmentSetup/InsertFeederCRDOffset

Input Paramters:

- [FromBody]
  - Feeder_ID– required, must be valid feeder ID
  - CRD
  - Offset
  - UserID_ID – required, must be valid user
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:
Success: Will return 1
Failure: Will return error message

Preconditions:

Note:

55. InsertSetup

Api/Method:

- EquipmentSetup/InsertSetup

Input Paramters:

- [FromBody]
    - MachineName
    - ProgramName
    - Assembly
    - TotalComponents
    - UserID_ID – required, must be valid user
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:
Success: Will return EquipmentSetup_ID
Failure: Will return error message

Preconditions:

Note:

56. InsertSetupSheetComment

Api/Method:

- EquipmentSetup/InsertSetupSheetComment

Input Paramters:

- [FromBody]
    - EquipmentSetup_ID
    - KeyDate
    - Comment
    - UserID_ID – required, must be valid user
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:
Success: Will return 1
Failure: Will return error message

Preconditions:

Note:

57. ListEquipmentTableComponentById

Api/Method:

- EquipmentSetup/ListEquipmentTableComponentById

Input Paramters:

- [FromBody]ListEquipmentTableComponentById
    - equipId
    - table – equipment table
    - feederTrayTrackId
    - langId – “0” is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
    - Equipment_ID
    - EquipmentMaster_ID
    - Make
    - Vendor
    - Model
    - CommonName
    - EquipmentTable
    - FeederTrayTrack_ID
    - Slot
    - Tray
    - Track
    - GRN_ID
    - GRN
    - Material_ID
    - Material
    - Qty
    - TransPort_ID
    - Transport
    - UserID_ID
    - LastUpdated
    - CheckDateQty

Note:

58. ListFeederByEquipmentSetup

Api/Method:

- EquipmentSetup/ListFeederByEquipmentSetup

Input Paramters:

- [FromBody]ListFeederByEquipmentSetup
  - setupId – equipmentsetup_id
  - langId – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - Feeder_ID
  - EquipmentSetup_ID
  - CommonEquipmentSetup_ID
  - CommonSetupName
  - Equipment_ID
  - Vendor
  - Model
  - CommonName
  - MachineName
  - ProgramName
  - LoadDateTime
  - Assembly_ID
  - Number
  - Revision
  - Version
  - AssemblyText
  - FeederTrayTrack_ID
  - Feeder
  - Tray
  - Track
  - FeederType
  - Media
  - Material_ID
  - Material
  - Descr
  - CRDCount
  - OutOfStock

- Deviation
- Change
- XFeeder
- LastUpdated
- CheckDate
- EffectiveDateFrom
- EffectiveDateTo

Note:

59. ListGRNTransportAssociationWithinTime

Api/Method:

- EquipmentSetup/ ListGRNTransportAssociationWithinTime

Input Paramters:

- [FromBody] ListGRNTransportAssociationWithinTime
  - startTime – required in datetime format
  - endTime – required in datetime format
  - maxCount – optional, default = 400
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - LoadTime
  - Transport
  - GRN
  - WindowsUserID

Note:

- The time between startTime and endTime is limited to 12 hours or 720 minutes

60. ListLoadedFeederByEquipment

Api/Method:

- EquipmentSetup/ListLoadedFeederByEquipment

Input Paramters:

- [FromBody]ListLoadedFeederByEquipment
  - equipId

- table – “0” if not specific
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - Transport
  - Transport_ID
  - Slot
  - Tray
  - Track
  - EquipmentTable
  - Qty
  - GRN_ID
  - GRN

Preconditions:

Note:

61. ListSetupByEquipmentAssembly

Api/Method:

- EquipmentSetup/ListSetupByEquipmentAssembly

Input Paramters:

- [FromBody]ListSetupByEquipmentAssembly
  - equipId
  - assemId
  - langId – “0” is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional
-

Expected Result:

- List Of
  - EquipmentSetup_ID
  - RFDisplay
  - SetupNumber
  - SetupVersion
  - MaxVersion
  - MachineName
  - ProgramName
  - ForceValidGRN
  - ProhibitLastPart

Note:

62. SetSetupTableActiveByLane

Api/Method:

- EquipmentSetup/SetSetupTableActiveByLane

Input Paramters:

- [FromBody]SetSetupTableActiveByLane
    - usrId – required and validated
    - setupId
    - equipId
    - activeTable
    - routeStepId
    - lane – “” if not specific
    - xOutId – “0” if not specific
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- (success: + xoutId)/error

Preconditions:

Note:

63. SetupValidation

Api/Method:

- EquipmentSetup/SetupValidation

Input Paramters:

- [FromBody]SetupValidation
    - setupId – equipmentsetup_id
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
    - Block

- ComponentLocation
- Tray
- Track
- RequiredPart
- ActualPart
- GRN – GRN++Number in sp
- Action

Preconditions:

Note:

64. UpdateEquipmentSetup

Api/Method:

- EquipmentSetup/UpdateEquipmentSetup

Input Paramters:

- [FromBody]
  - EquipmentSetup_ID– required, must be valid equipment setup ID
  - RouteStep_ID
  - Equipment_ID
  - Assembly_ID
  - SetupNumber
  - SetupVersion
  - TotalComponents
  - TotalPins
  - Export
  - Active
  - RFDisplay
  - CommonEquipmentSetup_I
  - UserID_ID – required, must be valid user
  - CheckDate – required, must be match time of last known update
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

Success: Will return a record with:

Result: Error code from stored procedure. 0 = success.

CheckDate: New checkdate of record.

Failure: Will return error message

Preconditions:

Note:

Label

65. EvaluateLabelFields

Api/Method:

- Label/EvaluateLabelFields

Input Paramters:

- [FromBody]EvaluateLabelFields
  - custId
  - usrId
  - lblType
  - assemId
  - fmaId
  - batchId
  - langId – “0” is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - FieldOrder
  - LabelFieldType_ID
  - LabelField
  - QuantityPerLabel
  - FieldValue

Preconditions:

Note:

66. GetContainerLabelsByContainer

Api/Method:

- Label/GetContainerLabelsByContainer

Input Paramters:

- [FromBody]GetContainerLabelsByContainer
    - custId
    - container
    - status
    - langId – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
    - ContainerCapacity
    - SerialNumber_ID
    - SequenceNumber
    - ContainerStatus_ID
    - LabelType_ID
    - LabelConfig_ID
    - AssemblySpecific
    - LabelPackage_ID
    - LabelGroup_ID
    - LabelType
    - Description
    - LabelQuantity
    - DuplicateQuantity
    - BatchID_ID

Preconditions:

Note:

67. ListLabelConfig

Api/Method:

- Label/ListLabelConfig

Input Paramters:

- [FromBody]ListLabelConfig
    - custId
    - lblType
    - assemId
    - langId – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- LabelType_ID
- LabelType
- Description
- AssemblySpecific
- SerialNumber_ID
- HasSerialNumberFields
- HasUserPrompts
- HasBatchPrompts
- HasMARoutePrompts
- LabelPackage_ID
- VerifyLabelCount
- ScanLabelCount
- LabelGroup_ID
- LabelDataDropPath
- LabelDefinitionPath
- LabelExecutablePath
- PrinterDefinitionPath
- LabelConfig_ID
- Assembly_ID
- DefinitionFile
- LabelStock
- RibbonStock
- DataFile
- Enabled
- BookEndLabel_ID
- UserID_ID
- LastUpdated
- CheckDate
- Number
- Revision
- Version
- AssemblyDesc
- BookEndLabelType

Preconditions:

Note:

68. PrintBoxLabel

Api/Method:

- Label/PrintBoxLabel

Input Paramters:

- [FromBody]PrintBoxLabel
    - custId
    - usrId
    - box
    - family
    - assem
    - rev
    - capacity
    - boxCnt
    - created
    - status
    - usrName
    - station
    - contRule
    - assemId
    - fmaId
    - batchId
    - lblPackageCode
    - printer
    - extraData – json string, e.g.,

```
[
    {
        "linkObject":"78-0000-01",
        "rev":"01",
        "linkData": "JMX1212121",
        "part":"",
        "partRev":""
    }
]
```

    - example
    - langId – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Note:

69. PrintLabel

Api/Method:

- Label/PrintLabel

Input Paramters:

- [FromBody]PrintLabel
    - custId
    - usrId
    - lblType
    - lblConfigId
    - qty
    - dupQty
    - exePath
    - dataDropPath
    - lblDefPath
    - assemId
    - fmaId
    - batchId
    - lblPackageCode
    - printer
    - lblDefFile
    - extraData – json string, e.g.,

```
[
    {
        "linkObject":"78-0000-01",
        "rev":"01",
        "linkData": "JMX1212121",
        "part":"",
        "partRev":""
    }
]
```

    - example
    - langId – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Note:

- json example

```
{
    "custId":"52",
    "usrId":"17328",
        "lblType":"FULLBOX",
        "lblConfigId":"34309",
        "qty":"1",
```

```
            “dupQty":"0",
            “exePath":"\\\\GDLLOFT02\\llmwin32\\LABELS",
            “dataDropPath":"c:\\Temp",
            “lblDefPath":"\\\\GDLLOFT02\\llmwin32\\LABELS",
            “assemId":"87941",
            “fmaId":"0",
            “batchId":"0",
            “lblPackageCode":"1",
            “printer":"",
            “dataFile":"",
            “lblDefFile":"ciscoempaquearrow.lwl",
            “example":"0",
            “langId":"0",
            “extraData":@"{""SOFTWAREREV"":"""",""FIRMWAREREV"":"""",""SERIE1"":""JAD2303
            0JTH"",""SERIE2"":""JAD23030JTH"",""BOXSERIALNUMBER"":""JMX854847"",""BOXCOUN
            T"":""2""}
        }
```

Link

70. GetBoardLinkInfo

Api/Method:

- Link/GetBoardLinkInfo

Input Paramters:

- [FromBody]GetBoardLinkInfo
  - wipId
  - linkObject
  - LinkObjectRev
  - langId – “0” is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- Wip_ID
- ChildWip_ID
- LinkData_ID
- LinkData
- LinkMaterial_ID
- LinkObject
- LinkObjectRevision
- UserID
- UserID_ID
- LastUpdated
- RouteStep_ID

- Equipment_ID

Preconditions:

Note:

71. LinkByLinkStation

Api/Method:

- Link/LinkByLinkStation

Input Paramters:

- [FromBody] LinkByLinkStation
  - custId – required and validated
  - serial – required
  - assemId
  - routeStepId
  - equipId
  - linkDataJson
  - usrId
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

Note:

- linkDataJson definition:
  - linkObject – linkObject configured on LinkStation
  - rev – linkObjectRev configured on LinkStation
  - linkData – linkData
  - part – scanned part number if alternate material is used. In the case when part == linkObject, linkData is for primarty material
  - partRev – part revision for alternate material. It is only used if linking alternate material (part != linkObject)
  - example:

```
[
    {
        "linkObject":"78-0000-01",
```

"rev":"01",
"linkData": "JMX1212121",
"part":"",
"partRev":""
},
{
"linkObject":"75-2222-03",
"rev":"03",
"linkData":"JMX3232323",
"part":"",
"partRev":""
}
]

- All or nothing linking for the entire link station
- Business rule implemented:
  - wip must belong to the customer and active
  - wip must of selected assembly or can be progress to selected assembly
  - wip must pass pv check via wip move
  - link data for all link objects configured on the route step are present
  - if ScanRev is turned on for a particular link object, revision must be supplied and validated
  - if unique, it must not be used in the system per material
  - if datamasks are configured, a link data is subjected to datamask verification
  - if alternate material (by supplying part different than linkobject in linkDataJson), alternate material will be verified
  - if linking a child board, the child assembly must be validated
  - if linking a child board, the child wip must belong to the customer and active
  - if linking a child board, pv of child wip
  - link nonunique component when applied
  - link unique component when applied
  - link child component when applied
  - wip move for child component when applied
  - wip move for parent wip when applied

72. LinkWithValidation

Api/Method:

- Link/LinkWithValidation

Input Paramters:

- [FromBody]LinkWithValidation
  - custId – required and validated

- assemId – assembly id of the board
    - deviationId – board deviation id. Pass in "0" if no deviation
    - boardSerial
    - linkObject
    - linkObjectRev
    - linkData
    - usrId – required and validated
    - routeStepId – required
    - equipId – required
    - part = "" – for future purpose, if scanned in part number is different than link object
    - rev = "" – for future purpose, if scanned in rev is different than link object rev
    - bypassChildLinkStepConfig = "0" – optional, only applies to the case when child board linking route step configuration is required
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

- Assembly must be configuration for Linking operation via LinkStation

Note:

- Linking operation:
    - Wip-In: Before linking is called, wip must move into route step that is configured for linking first. This is done by calling Wip/BoardWipMove (with IOType_ID =1, OpStatus = 4)
    - Client will perform the linking of all required materials by calling this LinkWithValidation() function
    - Wip-Out: Upon completion of all required materials' linking, call Wip/BoardWipMove (with IOType_ID =2, OpStatus = 4) to signal the completion of the linking
    - Wip-Out: call Wip/BoardWipMove (with IOType_ID =2, OpStatus = 3) in the case of error to signal the aborting of the operation
- This is a combined function to do the following
    - Validate board and its assembly id, as well as its availability for linking (Scrap, OnHold, Active, etc)
    - Get link station material and related information, including ScanPart, ScanRev, Serialize, Unique, ChildAssembly, AlternateMaterials, and Datamasks based on the board serial, board assembly id, board deviation id, and route step
    - Validate linkobject, linkobjectrev, and routestep to make the link object was configured and still needed on the route step
    - Validate link data's serial number, uniqueness, datamask if so configured.
    - Wip move for the child board if applied
    - Linking – record linkdata and board relationship in database.

- Child link
- Link unique serial
- Link non-unique serial

73. LinkWithValidationByText

Api/Method:

- Link/LinkWithValidationByText

Input Paramters:

- [FromBody]LinkWithValidationByText
  - customer – customer name; required
  - division – required
  - assembly – assembly number
  - revision – assembly revision
  - deviationId – board deviation id. Pass in “0” if no deviation
  - boardSerial
  - linkObject
  - linkObjectRev
  - linkData
  - usrId – required and validated
  - factory – required
  - ma – manufactory area; required
  - route – required
  - stepDescr – route step descr; required
  - equipment
  - part = "" – for future purpose, if scanned in part number is different than link object
  - rev = "" – for future purpose, if scanned in rev is different than link object rev
  - bypassChildLinkStepConfig = "0" – optional, only applies to the case when child board linking route step configuration is required
- sqlServer = "" – optional
- dataBase = "" – optional

string customer, string division, string assembly, string revision,

Expected Result:

- success/error

Preconditions:

- Assembly must be configuration for Linking operation via LinkStation

Note:

- Linking operation:

- Wip-In: Before linking is called, wip must move into route step that is configured for linking first. This is done by calling Wip/BoardWipMove (with IOType_ID =1, OpStatus = 4)
- Client will perform the linking of all required materials by calling this LinkWithValidation() function
- Wip-Out: Upon completion of all required materials' linking, call Wip/BoardWipMove (with IOType_ID =2, OpStatus = 4) to signal the completion of the linking
- Wip-Out: call Wip/BoardWipMove (with IOType_ID =2, OpStatus = 3) in the case of error to signal the aborting of the operation

- This is a combined function to do the following
  - Validate board and its assembly id, as well as its availability for linking (Scrap, OnHold, Active, etc)
  - Get link station material and related information, including ScanPart, ScanRev, Serialize, Unique, ChildAssembly, AlternateMaterials, and Datamasks based on the board serial, board assembly id, board deviation id, and route step
  - Validate linkobject, linkobjectrev, and routestep to make the link object was configured and still needed on the route step
  - Validate link data's serial number, uniqueness, datamask if so configured.
  - Wip move for the child board if applied
  - Linking – record linkdata and board relationship in database.
    - Child link
    - Link unique serial
    - Link non-unique serial

Material

74. BlockGRN

Api/Method:

- Material/BlockGRN

Input Paramters:

- [FromBody]BlockGRN
  - grn – required.
  - blockTypeId – required.
  - blocked – required;
  - assembly – optional
  - revision – optional
  - version – optional
  - family – optional

- mfgArea – optional
    - route – optional
    - usrId – required.
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

- GRN must not be loaded in a setup sheet.

Note:

- Provide the corrects parameters depending on the BlockType_ID. For example, if BlockType_ID is 5, add in the parameters the name of family.

75. CheckGRNMSL

Api/Method:

- Material/CheckGRNMSL

Input Paramters:

- [FromBody]CheckGRNMSL
    - usrId
    - grnId
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- integer value of ElapsedExposureTime

76. GetGRNByName

Api/Method:

- Material/GetGRNByName

Input Paramters:

- [FromBody]GetGRNByName
    - grn
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- GRN_ID
- GRN
- MatDocNumber
- DateCode
- SeqNumber
- Material_ID
- Material
- Descr
- Vendor
- ReelIsEmpty
- SplitReelParent
- SplitReelChild
- MultiPartParent
- MultiPartChild
- UserID_ID
- LastUpdated
- CheckDate
- SAPDateCode
- SAPLotCode
- ElapsedExposureTime
- BagOpened
- BagClosed
- BagStatus
- BakeIn
- BakeOut
- BakeStatus
- BakeTemperature

Note:

77. GetGRNExposureTime

Api/Method:

- Material/GetGRNExposureTime

Input Paramters:

- [FromBody]UpdateGRNBagStatus
    - grnId
- sqlServer = "" – optional

- dataBase = "" – optional

Expected Result:

- ExposureTime – in seconds

Preconditions:

Note:

78. GetMaterialByName

Api/Method:

- Material/GetMaterialByName

Input Paramters:

- [FromBody]
  - Material
  - Language_ID– “0” is not localized or English – optional

- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- Material_ID
- Material
- Description
- MRPController
- PartType
- ProcType
- SprocType
- PinCount
- CompoentType
- BulkIssueFlag
- Assembly
- Phantom
- UserID_Id
- CheckDate

Preconditions:

Note:

79. GetMSDExposureByGRN

Api/Method:

- Material/GetMSDExposureByGRN

Input Paramters:

- [FromBody]GetMSDExposureByGRN
  - grn
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- GRN
- BagOpened
- BagClosed
- BagStatus
- ElapsedExposureTime
- Material
- ExposureLevel_ID
- ExposureTime

Note:

80. InsertGRNBagHistory

Api/Method:

- Material/InsertGRNBagHistory

Input Paramters:

- [FromBody]InsertGRNBagHistory
  - usrId – required and validated
  - grnId – required and validated
  - grn
  - bagOpened – datetime of bag opening; can be “”
  - bagClosed – datetime of bag closing; can be “”
  - bagStatus – numeric value (sql int)
  - elapsedTime – numeric value (sql int)
  - updateMode – 0, 1, or 2
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Note:

- UpdateMode
  - 0 = Opened
    - Set bagClosed to “” regardless input parameter
  - 1 = Closed
  - 2 = ReOpened
    - Set bagClosed to “”
- Insert a record in CR_GRNBagHistory

81. InsertGRNBakeHistory

Api/Method:

- Material/InsertGRNBakeHistory

Input Paramters:

- [FromBody]InsertGRNBakeHistory
  - usrId – required and validated
  - grnId – required and validated
  - grn
  - bakeIn – datetime of bake in; can be “”
  - bakeOut – datetime of bake out; can be “”
  - bakeStatus – numeric value (sql int), 1 or other
  - elapsedTime – numeric value (sql int)
  - bakeCompleted – numeric value, 1 or other
  - bakeTemperature – numeric value (sql int)
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

Note:

- bakeStatus
  - 1 = bakIn
    - Set bakeOut to “” regardless input parameter
  - Otherwise
    - If bakeCompleted = 1
      - bakeStatus = 0

        - else
            - bakeStatus = 0
            - bakeout = "1900-01-01 00:00:00.000"
- Insert a record in CR_GRNBakeHistory

82. UnblockGRN

Api/Method:

- material/UnblockGRN

Input Paramters:

- [FromBody]BlockGRN
    - grn – required.
    - blockTypeId – required.
    - blocked – required.
    - assembly – optional
    - revision – optional
    - version – optional
    - family – optional
    - mfgArea – optional
    - route – optional
    - usrId – required.
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

- GRN must be previously blocked.

Note:

- Provide the corrects parameters depending on the BlockType_ID. For example, if BlockType_ID is 5, add in the parameters the name of family.

83. UpdateGRNBagStatus

Api/Method:

- Material/UpdateGRNBagStatus

Input Paramters:

- [FromBody]UpdateGRNBagStatus
  - usrId
  - grnId
  - bagOpened
  - bagClosed
  - bagStatus – Open or Close
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Note:

- bagStatus input
  - Open
    - To close the bag
    - Set BagClosed to db time with offset regardless input parameter
    - ElapsedExposureTime = GRN BagExposureTime – Time difference between BagOpened and BagClosed in mins
    - bakeStatus = 01
  - Close
    - If BagOpened = '1900-01-01 00:00:00.000'
      - To open the bag
      - bagStatus = 1
      - ElapsedExposureTime = GRN BagExposureTime
    - Otherwise
      - This is a reopen case
      - Set BagOpened to db time with offset regardless input parameter
      - Set BagClosed = '1900-01-01 00:00:00.000'
      - bagStatus = 1
      - ElapsedExposureTime = GRN BagExposureTime
- Update CR_GRNs table for the BagOpened/BagClosed, ElapsedExposureTime, and BagStatus
- Insert a record in CR_GRNBagHistory

84. UpdateGRNBagStatusByGRN

Api/Method:

- Material/UpdateGRNBagStatusByGRN

Input Paramters:

- [FromBody]UpdateGRNBagStatus
  - usrId
  - grn
  - updateTime – time to open or close grnBag
  - bagStatus – Open or Close grn bag
  - elapsedTime – optional; set to -1 if not updating GRN's ElapsedExposureTime
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Note:

- Open bag
  - Update CR_GRNs table for the GRN with
    - bagOpened = updateTime
    - bagClosed -- set to '1900-01-01 00:00:00.000' because until it is close it has the full exposure....
    - bagStatus = 1
    - ElapsedExposureTime = GRN exposureTime if elapsedTime is not null or -1.
  - record inserted in CR_GRNBagHistory
    - bagOpened = updateTime
    - bagClosed = '1900-01-01 00:00:00.000'
    - bagStatus = 1
    - ElapsedExposureTime = GRN exposureTime by configuration
- Close bag
  - Update CR_GRNs table for the GRN with
    - bagOpened -- no change
    - bagClosed = updateTime
    - bagStatus = 0
    - ElapsedExposureTime = no change
  - record inserted in CR_GRNBagHistory
    - bagOpened -- no change
    - bagClosed = updateTime
    - bagStatus = 0
    - ElapsedExposureTime = GRN BagExposureTime – Time difference between BagOpened and BagClosed in mins

85. UpdateGRNBakeStatus

Api/Method:

- Material/UpdateGRNBakeStatus

Input Paramters:

- [FromBody]UpdateGRNBakeStatus
  - usrId – required and validated
  - grnId – required and validated
  - grn
  - bakeIn – datetime of bake in; can be “”
  - bakeOut – datetime of bake out; can be “”
  - bakeStatus – “”, “Bake In”, or “Bake Out”
  - elapsedTime – numeric value (sql int)
  - bakeCompleted – numeric value, 1 or other
  - bakeTemperature – numeric value (sql int); if “”, default to 0
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

Note:

- bakeStatus input
  - “” or “Bake Out”
    - Set bakeIn to db time with offset regardless input parameter
    - bakeOut = "1900-01-01 00:00:00.000"
    - bakeStatus = 1
  - Otherwise
    - If “Bake In”
      - bakeStatus = 0
      - if bakeCompleted = 1
        - bakeOut = db time with offset
        - elapsedTime = GRNExposureTime*60
      - otherwise,
        - bakeout = "1900-01-01 00:00:00.000"
- Update CR_GRNs table for the bakeIn/BakeOut, BakeStatus, ElapsedExposureTime, and BakeTemperature
- Insert a record in CR_GRNBakeHistory

Panel

86. BreakoutPanel

Api/Method:

- Panel/BreakupPanel

Input Paramters:

- [FromBody]BreakupPanel
  - usrId – required and validated
  - serial – wip serial that is on the panel
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

- all serials on the panel must have been birthed via other MES module before it can be broken.

Note:

- only set BoardPanelBroken == 1 if wips on the panel are not Scrapped or Purged, and stil attached to the panel

87. GetPanelById

Api/Method:

- Panel/GetPanelById

Input Paramters:

- [FromBody]GetPanelById
  - panelId
  - langId – "0" is not localized or English
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - Panel_ID
  - Panel
  - PanelName
  - BoardsPerPanel
  - Rows
  - Columns
  - NumberOfSides
  - SerialNumberType

- UserID_ID
- LastUpdated
- CheckDate

Preconditions:

Note:

88. GetPanelByWipId

Api/Method:

- Panel/GetPanelByWipID

Input Paramters:

- [FromBody]GetPanelByWipId
  - usrId
  - wipId
  - langId – “0” is not localized or English
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - Panel_ID
  - Panel
  - WIP_ID
  - SerialNumber
  - Mapping
  - XOut
  - XOutStats

Preconditions:

Note:

89. ListConnectedSerialInPanel

Api/Method:

- Panel/ListConnectedSerialInPanel

Input Paramters:

- [FromBody]ListConnectedSerialInPanel
    - panelId
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
    - Panel_ID
    - Panel
    - WIP_ID
    - SerialNumber
    - Mapping
    - XOut
    - XOutStats

Preconditions:

Note:

90. ListConnectedSerialInPanelWithCoordinates

Api/Method:

- Panel/ListConnectedSerialInPanelWithCoordinates

Input Paramters:

- [FromBody]ListConnectedSerialInPanelWithCoordinates
    - panelId
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
    - Panel_ID
    - Panel
    - WIP_ID
    - SerialNumber
    - Mapping
    - XOut
    - XOutStats
    - BoardRow
    - BoardColumn

Preconditions:

Note:

91. ListPanelMappingById

Api/Method:

- Panel/ListPanelMappingById

Input Paramters:

- [FromBody]ListPanelMappingById
  - panelId
  - langId – “0” is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - Panel_ID
  - Panel
  - PanelText
  - Mapping
  - BoardRow
  - BoardColumn
  - UserID_ID
  - LastUpdated
  - CheckDate

Preconditions:

Note:

92. ListPanelSerialByWipId

Api/Method:

- Panel/ListPanelSerialByWipId

Input Paramters:

- [FromBody]ListPanelSerialByWipId
  - wipId
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
    - Panel_ID
    - Panel
    - WIP_ID
    - SerialNumberOri
    - Mapping
    - XOut
    - XOutStats
    - SerialNumber

Preconditions:

Note:

PV

93. ProcessVerify

Api/Method:

- PV/ProcessVerify

Input Paramters:

- [[FromBody] AgingWipsReportByCustomer
    - custId
    - usrId
    - serial
    - chkpt – checkpoint name; can be “”
    - assemId
    - assem
    - rev
    - procAsPanel
    - langId
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

Note:

Reporting

94. AgingWipsReportByCustomer

Api/Method:

- Reporting/AgingWipsReportByCustomer

Input Paramters:

- [[FromBody] AgingWipsReportByCustomer
    - custId
- daysEscaped – EndTime in WP_CurrentWipRouteStep for the wip that is up-to this number of days from current datetime.
    - langId
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
    - SerialNumber – wip serial number
    - IsRMA – whether the wip is a RMA
    - RMAReturnCount – # of time this wip has returned as RMA
    - Batch – batch associated with specific route step move
    - FamilyText
    - Assembly_ID
    - AssemblyNumber
    - AssemblyRevision
    - Route
    - CurrentStep
    - CurrentRouteStep
    - LastWipMoveTime – last datetime in WP_AssemblyRouteWip for this wip/routestep
    - AgingDays – days since LastWipMoveTime (above)

Preconditions:

Note:

- Please work with your application admin to make sure sure the web.config of api have the following setting:
    - configuration setting, sqlReportingTimeout, is used to alter SQL command timeout for GetReport method
        - Timeout in seconds.
        - E.g. 300. Please note that 0 will not be accepted. In the case 0 is set, the api will assume default behavior of 30 secs timeout

95. GetReport

Api/Method:

- Reporting/GetReport

Input Paramters:

- [FromBody]GetReport
  - rptSPName
  - inputJson
  - usrId
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- return of customized stored procedure as “rptSPName”

Preconditions:

Note:

- please refer to document, “ MESWebApi – writing customized sp to get reporting data via generic api call ” for implementation details

Route

96. ListRouteAssemblyByCustomer

Api/Method:

- Route/ListRouteAssemblyByCustomer

Input Paramters:

- [FromBody]ListRouteAssemblyByCustomer
  - customerId – string[], e.g., [2,3]
  - routeStatus – e.g., “Active”
  - updatedAfter – e.g., “2019-07-01”
  - langId – “0” is not localized or English
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
    - CustomerName
    - DivisionName
    - AssemblyNumber
    - Route
    - AssemblyRevision
    - RouteStatus
    - UpdatedDate

Preconditions:

Note:

97. ListRouteAssemblyByCustomerId

Api/Method:

- Route/ListRouteAssemblyByCustomer

Input Paramters:

- [FromBody]ListRouteAssemblyByCustomer
    - custId – valid customer_ID
    - activeRouteOnly – 0 or 1; default to 1
    - updatedAfter – if null, return data changed in past 30 days
    - langId – "0" is not localized or English
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
    - Customer_ID
    - CustomerName
    - DivisionName
    - AssemblyNumber
    - AssemblyRevision
    - Route
    - RouteStatus
    - UpdatedDate

Preconditions:

Note:

98. ListRouteStep

Api/Method:

- Route/ListRouteStep

Input Paramters:

- [FromBody]ListRouteStepByFactoryMARoute
  - factory – “” if don’t want to specify factoryname
  - langId – “0” is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
  - RouteStep_ID
  - FactoryMARoute_ID
  - FactoryName
  - ManufacturingAreaName
  - RouteName
  - Step_ID
  - StepName
  - Descr
  - Description
  - Occurrence
  - StepOrder
  - XCoordinate
  - YCoordinate
  - StepType
  - StepTypeName
  - NextStep_ID
  - RouteValidation
  - BirthingStation
  - BackFlush
  - UseJUID
  - WorkCenter_ID
  - WorkCenterText
  - UserID_ID
  - LastUpdated
  - CheckDate

Preconditions:

Note:

99. ListRouteStepByFactoryMARoute

Api/Method:

- Route/ListRouteStepByFactoryMARoute

Input Paramters:

- [FromBody]ListRouteStepByFactoryMARoute
    - fmaRouteId – required and validated
    - step
    - langId – “0” is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
    - RouteStep_ID
    - StepName
    - Descr
    - Description
    - Occurrence
    - StepOrder
    - StepType,
    - XCoordinate
    - YCoordinate
    - Step_ID
    - WorkCenter_ID
    - StepType_IDte

Preconditions:

Note:

100. ListRouteStepByStepEquipmentAssembly

Api/Method:

- Route/ListRouteStepByStepEquipmentAssembly

Input Paramters:

- [FromBody]ListRouteStepByStepEquipmentAssembly
    - stepInstance
    - equipId
    - assem – assembly number
    - rev – assembly revision

- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
  - RouteStep_ID
  - FactoryText
  - MAText
  - RouteText
  - StepText
  - DescrText

Preconditions:

Note:

101. ListSubWorkCenter

Api/Method:

- Route/ ListSubWorkCenter

Input Paramters:

- [FromBody] `SubWorkCenterParameter`
  - stepInstance
  - Customer
  - Division
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- Customer
- Division
- ManufacturingArea
- ManufacturingAreaId
- Route
- RouteId
- Step
- StepId
- StepInstance
- StepInstanceId
- Equipment
- EquipmentId
- PassFail

102. ListProcessLink

Api/Method:

- Route/ListProcessLink

Input Paramters:

- [FromBody]`ProcessLinkRequestParameter`
    - Customer
    - Division
    - AssemblyNumber
    - AssemblyRevision
    - StepInstance
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- Customer
- Division
- Assembly
- Revision
- ManufacturingArea
- ManufacturingAreaId
- Route
- RouteId
- Step
- StepId
- StepInstance
- StepInstanceId
- Equipment
- EquipmentId
- PassFail

RouteStepSetup

103. GetRouteStepComponetUseBin

Api/Method:

- RouteStepSetup/GetRouteStepComponentUseBin

Input Paramters:

- [FromBody]GetRouteStepComponentUseBin
  - routeStepId
  - bin
  - grnId,
  - loadTime
  - langId – “0” is not localized or English
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- RouteStep_ID
- FactoryText
- MAText
- RouteText
- StepText
- Bin
- Feeder
- Tray
- Track
- GRN_ID
- GRN
- Material
- LoadTime
- LoadUserID_ID
- RemovalTime
- RemovalUserID_ID
- Qty
- RouteStepSetup_IDs

Preconditions:

Note:

104. ListActiveRouteStepSetupByRouteStepAssembly

Api/Method:

- RouteStepSetup/ListActiveRouteStepSetupByRouteStepAssembly

Input Paramters:

- [FromBody]ListActiveRouteStepSetupByRouteStepAssembly
  - routeStepId
  - assemId
  - langId – “0” is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - RouteStepSetup_ID
  - Assembly_ID
  - Number
  - Revision
  - Version
  - AssemblyName
  - BOM_ID
  - RouteStep_ID
  - FactoryName
  - FactoryMAName
  - RouteName
  - StepName
  - SetupNumber
  - SetupVersion
  - TotalComponents
  - TotalPins
  - Alias
  - Export
  - Active
  - RFDisplay
  - MachineName
  - ProgramName
  - LoadDateTime
  - Assembly
  - CommonRouteStepSetup_ID
  - CommonSetupName
  - UserID_ID
  - LastUpdated
  - CheckDate

Preconditions:

Note:

105. ListRouteStepComponentById

Api/Method:

- RouteStepSetup/ListRouteStepComponentById

Input Paramters:

- [FromBody]ListRouteStepComponentById
  - routeStepId
  - bin
  - langId – “0” is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - RouteStep_ID
  - FactoryName
  - ManufacturingAreaName
  - RouteName
  - StepName
  - Bin
  - Feeder
  - Tray
  - Track
  - GRN_ID
  - GRN
  - Material
  - Qty
  - UserID_ID
  - LastUpdated
  - CheckDateQty

Preconditions:

Note:

106. ListRouteStepSetupBinBySetup

Api/Method:

- RouteStepSetup/ListRouteStepSetupBinBySetup

Input Paramters:

- [FromBody]ListRouteStepSetupBinBySetup
    - setupId – routestepsetup_id
    - langId – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
    - RouteStepSetupBin_ID
    - RouteStepSetup_ID
    - CommonRouteStepSetup_ID
    - CommonSetupName
    - Assembly_ID
    - Number
    - Revision
    - Version
    - AssemblyName
    - RouteStep_ID
    - FactoryName
    - ManufacturingAreaName
    - RouteName
    - StepName
    - Bin
    - Feeder
    - Tray
    - Track
    - Material_ID
    - Material
    - Qty
    - OutOfStock
    - Deviation
    - UserID_ID
    - LastUpdated
    - CheckDateQty

Preconditions:

Note:

107. RouteStepComponentAddPart

Api/Method:

- RouteStepSetup/RouteStepComponentAddPart

Input Paramters:

- [FromBody]RouteStepComponentAddPart
  - usrId – required and validated
  - setupId – routestepsetup_ID
  - routeId
  - bin
  - grnId
  - qty
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

Note:

108. RouteStepComponentRemovePart

Api/Method:

- RouteStepSetup/RouteStepComponentRemovePart

Input Paramters:

- [FromBody]RouteStepComponentRemovePart
  - usrId – required and validated
  - routeStepId
  - bin,
  - grnId
  - reelIsEmpty
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

Note:

109. RouteStepSetupValidation

Api/Method:

- RouteStepSetup/RouteStepSetupValidation

Input Paramters:

- [FromBody]RouteStepSetupValidation
    - setupId – routeStepSetup_ID
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

Note:

Security

110. GetUserByUserId

Api/Method:

- Security/GetUserByUserId

Input Paramters:

- [FromBody]GetUserByUserId
    - usrId – UserID_ID
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- UserID_ID, UserID, WindowsUserID, LastName, FirstName, Active

Preconditions:

- Valid and active MES user

Note:

111. GetUserByWindowsId

Api/Method:

- Security/GetUserByWindowsId

Input Paramters:

- [FromBody]GetUserByWindowsId
    - user – windows NT userId; if service account is used, it will be the name of the service account.
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- UserID_ID, UserID, WindowsUserID, LastName, FirstName, Active

Preconditions:

- Valid and active MES user

Note:

- This is the first api to call before each session. Client should cache UserID_ID in the front end. Most of "wirte" api's (insert or update database) will need to pass in this parameter (usrId)
- Return UserID_ID for active user only

112. ListGroupsByUserId

Api/Method:

- Security/ListGroupsByUserId

Input Paramters:

- [FromBody]
    - UserID_ID
    - PartialKey – use this to narrow the groups that are returnned
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of

- Group_ID
- GroupDescr – Security group name
- UserID_ID
- UserId
- LastName
- FirstName
- AuditUserID_ID
- CheckDate

Preconditions:

Note:

Serial

113. GenerateSerialNumbers

Api/Method:

- Serial/GenerateSerialNumbers

Input Paramters:

- [FromBody]GenerateSerialNumbers
  - custId – required and validated
  - usrId – required and validated
  - serialName – SerialNumberName in MES
  - qty – 1+
  - assemId – Assembly_ID for the serial number
  - fmaId – FactoryMA_ID
  - batchId – valid or 0
  - example = "0" if not testing/sampling
  - langId – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of (RowNumber, SerialNumber)

Preconditions:

- Use MES to configure serialnumber

Note:

- If example is set to "0" (default), serial numbers will be generated and inserted into SerialNumberLog table.

114. GetBarcodeMaskById

Api/Method:

- Serial/GetBarcodeMaskById

Input Paramters:

- [FromBody]GetBarcodeMaskById
  - barcodeMaskId
  - langId – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- BarcodeMask_ID
- BarcodeMask
- BarcodeMaskName
- Mask
- UserID_ID
- LastUpdated
- CheckDate

Preconditions:

Note:

Site

115. ListCustomer

Api/Method:

- site/ListCustomer

Input Paramters:

- [FromBody]ListCustomer
  - partialKey – required
  - active = "1" if active only
  - langId – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - Customer_ID
  - Customer
  - CustomerName
  - Division
  - DivisionName
  - CustomerDivisionName
  - SAPIdentifier
  - ForceValidGRN
  - ProhibitLastPart
  - RequireQuickscan
  - UserID_ID
  - System.DateTime LastUpdated
  - CheckDate
  - Active

Preconditions:

Note:

- To get active customer, set active = “1”.

Test

116. GetCurrentRouteStep

Api/Method:

- *Test/*GetCurrentRouteStep

Input Paramters:

- [FromBody]GetCurrentRouteStep
  - serialNumber – required

- sqlServer = "" – optional
- dataBase = "" – optional

117. GetLastTestResult

Api/Method:

- *Test/*GetLastTestResult

Input Paramters:

- [FromBody]GetAllHistoryResult
  - serialNumber -required
  - customerName -required
  - division
  - processStep -required

- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
  - StartTime
  - StopTime
  - TestStatus
  - MachineName
  - Operator
  - StepOrTestName
  - FailureLabel
  - FailureMessage

118. GetPanelSerializeResult

Api/Method:

- Test/GetPanelSerializeResult

Input Paramters:

- [FromBody]GetPanelSerializeResult
  - customerName
  - division
  - panelSerialNumber - required
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - Panel_ID
  - Panel
  - WIP_ID
  - SerialNumberOri

- Mapping
- XOut
- XOutStats
- SerialNumber

119. GetProcessStepFailLoops

Api/Method:

- *Test/*GetProcessStepFailLoops

Input Paramters:

- [FromBody]GetProcessStepFailLoops
  - customerName
  - division - required
  - serialNumber
  - processStep
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- ConsecutiveFailCount – consecutive fail count since last “Pass” or absent at the processStep
- MaxLoopCount – max loop count configured for processStep
- LastOpStatus – last process status for the wip
- LastProcessStep – last process step name

120. GetTestDataFormats

Api/Method:

- *Test/*GetTestDataFormats

Input Paramters:

- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- String
    - TypeFlag Value

121. GetTestHistory

Api/Method:

- Test/GetTestHistory

Input Paramters:

- [FromBody]GetAllHistoryResult
    - serialNumber -required
    - customerName -required
    - division
    - processStep -required

- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
    - StartTime
    - StopTime
    - TestStatus
    - MachineName
    - Operator
    - StepOrTestName
    - FailureLabel
    - FailureMessage

122. GetTestHistoryWithDefects

Api/Method:

- Test/GetTestHistoryWithDefects

Input Paramters:

- [FromBody]
    - serialNumber -required
    - customerName -required
    - division
    - langId - “0” is not localized or English – optional

- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of:
    - StartTime
    - StopTime
    - TestStatus
    - MachineName
    - Operator
    - StepOrTestName
    - Process_ID
    - Equipment_ID
    - WIP_ID
    - RouteStep_ID
    - Assembly_ID
    - AssemblyNumber
    - Assembly
    - FactoryMARoute_ID
    - FactoryText
    - MAText
    - RouteText
    - Test_Process
    - Family_ID
    - FamilyText
    - FailureLabels: Null, or List of:
        - Data_ID
        - FailureLabel
        - FailureMessage
        - Analysis: null, or List of:
            - Analysis_ID
            - AnalysisDateTime
            - DefectLocation
            - Defect_ID
            - Defect
            - DefectCategory_ID
            - DefectCategory
            - AnalysisEquipmentOrRouteStepSetup_ID

        - AnalysisEquipmentOrRouteStepSetup
        - AnalysisIsEquipmentSetup
        - Material_ID
        - Material
        - DefectCount
        - FeederTrayTrack_ID
        - DefectDetail
        - AnalysisRouteStep_ID
        - AnalysisRouteStep
        - AnalysisOperator
        - Repairs: Null, or List of:
            - Repair_ID
            - RepairDateTime
            - RepairCategory_ID
            - RepairCategory
            - RepairDetail
            - RepairMemo
            - RepairRouteStep_ID
            - RepairRouteStep
            - RepairOperator

Note:

Assembly = MES assembly description

123. InsertAnalysis

Api/Method:

- Test/InsertAnalysis

Input Paramters:

- [FromBody]
    - WIP_ID
    - Process_ID
    - Data_ID
    - langId– "0" is not localized or English – optional
    - userId – optional, defaults to 1 (system account) if not used
    - AnalysisRouteStep_ID – optional, see notes on assigned route step
    - Analysis_Factory – optional, see notes on assigned route step
    - Analysis_MA – optional, see notes on assigned route step
    - Analysis_Route – optional, see notes on assigned route step
    - Analysis_Step – optional, see notes on assigned route step
    - Analysis_StepInstance – optional, see notes on assigned route step
    - analysis

- Single record with:
  - defect_category (note: currently not used, pass in “”)
  - cause
  - crd
  - part_number
  - details
  - status (note: currently not used, pass in “”)
  - pin_count
  - comments
  - failure_label_id (note: currently not used, pass in 0)
  - test_event_id (note: currently not used, pass in 0)
  - work_station (note: currently not used, pass in “”)
  - isequipment (optional, see notes on feeder / bins)
  - equipmentorroutestep_id (optional, see notes on feeder / bins)
  - feedertraytrack_id (optional, see notes on feeder / bins)
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- Analysis_ID

Note:

The following fields are verified and should match a value in MES

analysis/cause – Should match a defect in MES
analysis/crd_part_number – Refers to a material, should match a material in MES

Notes on selecting route step: By default, the analysis is assigned to the same route step as the test being referenced. You can override this by passing in one of two values:

AnalysisRouteStep_ID: Points to the exact route step ID you wish to reference

Analysis_Factory, Analysis_MA, Analysis_Route, Analysis_Step, Analysis_StepInstance: This will query the route based on text passed in for factory, manufacturing area (“MA”), route, step, and step instance / description. If the route step ID cannot be found, an error will return.

If both these values are passed in, AnalysisRouteStep_ID will override the text lookup parameters.

Notes on selecting feeder / bins: By default, the MES Web API will try to compute the equipment or route setup and feeder / bin based on the CRD and WIP passed in. The web API will take the first result that matches. If this cannot be computed, this will be left blank (0 IDs) when writing the analysis record.

Optionally, you can pass in these values directly to the API.

For equipment setup / feeders, "EquipmentOrRouteStep_ID" should be set to the EquipmentSetup_ID of the equipment setup, "IsEquipment" should be 1, and "FeederTrayTrack_ID" should point to the ID of the feeder / tray / track where the CRD is.

If you are passing in route step setup / bins, "EquipmentOrRouteStep_ID" should be set to the RouteStepSetup_ID of the route step setup, "IsEquipment" should be 0, and "FeederTrayTrack_ID" should be the Bin value (found in CT_RouteStepSetupBins) where the CRD is.

- **InsertRework**

Api/Method:
- Test/InsertRework

Input Paramters:
- [FromBody]
    - WIP_ID
    - Process_ID
    - Data_ID
    - langId– "0" is not localized or English – optional
    - Analysis_ID
    - userId – optional, defaults to 1 (system account) if not used
    - ReworkRouteStep_ID – optional, see notes on assigned route step
    - Rework_Factory – optional, see notes on assigned route step
    - Rework _MA – optional, see notes on assigned route step
    - Rework _Route – optional, see notes on assigned route step
    - Rework_Step – optional, see notes on assigned route step
    - Rework_StepInstance – optional, see notes on assigned route step
    - rework
        - Single record (optional) with:
            - defect_id (note: currently not used, pass in 0)
            - rework_category
            - rework_detail
            - grn_number (optional, pass in "" if not applicable)
            - part_number (optional, pass in "" if not applicable)
            - status (note: currently not used, pass in "")
            - comments
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:
- 1 (or error message)

Note:

The following fields are verified and should match a value in MES

rework/rework_category – Should match a repair / rework category in MES
rework/grn_number – Should match a GRN number in MES
rework/part_number – Should match a material in MES

Notes on selecting route step: By default, the rework is assigned to the same route step as the test being referenced. You can override this by passing in one of two values:

ReworkRouteStep_ID: Points to the exact route step ID you wish to reference

Rework_Factory, Rework_MA, Rework_Route, Rework_Step, Rework_StepInstance: This will query the route based on text passed in for factory, manufacturing area ("MA"), route, step, and step instance / description. If the route step ID cannot be found, an error will return.

If both these values are passed in, ReworkRouteStep_ID will override the text lookup parameters.

124. ListDefects

Api/Method:

- Test/ListDefects

Input Paramters:

- [FromBody]
    - UserID_ID – Must be an active user ID in MES
    - PartialKey – Filters defect list
    - ActiveOnly – "true" or "false"
    - MaxCount – Optional, use 0 to retrieve everything
    - Language_ID – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
    - Defect_ID
    - Defect
    - DefectText
    - DefectCategory_ID
    - DefectCategory
    - Active

Note:

Returns a list of the current defect options, and their categories, in MES.

125. ListReworkCategories

Api/Method:
- Test/ListReworkCategories

Input Paramters:
- [FromBody]
  - partialKey – Filters rework list
  - maxCount – Optional, use 0 to retrieve everything
  - langId – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:
- List Of
  - RepairCategory_ID
  - RepairCategoryText
  - RequiresMRBLabel
  - EnableGRNField

Note:

126. ListTestDataWithinTime

Api/Method:
- Test/ListTestDataWithinTime

Input Paramters:
- [FromBody]ListTestDataWithinTime
  - startTime – required in datetime format
  - endTime – required in datetime format
  - maxCount – optional, default = 400
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:
- List of
  - ExtractionStartTime
  - ExtractionEndTime
  - Customer
  - Division
  - Assembly
  - Revision

- RouteText
- StepText
- DescrText
- SerialNumber
- TestStatus
- ProcessLoop
- StartDateTime
- StopDateTime

Preconditions:

Note:

- The time between startTime and endTime is limited to 30 minutes

127. OKToTest

Api/Method

- Test/OKToTest

Input Paramters:

- [FromBody]OKToTest
  - customerName
  - division - required
  - serialNumber
  - assemblyNumber
  - testerName - required
  - processStep
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- String
  - PASS/FAIL

128. OKToTestLinkMaterial

Api/Method:

- *Test/*OKToTestLinkMaterial

Input Paramters:

- [FromBody] FromBody]OKToTestLinkMaterial
    - customerName– required
    - Division
    - linkMaterialSerialNumber - required
    - sssemblyNumber
    - testerName - required
    - processStep
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- String
    - PASS/FAIL

129. OkToTest_Breakout

Api/Method:

- *Test/*OkToTest_Breakout

Input Paramters:

- [FromBody]OkToTest_Breakout
    - customerName
    - division
    - serialNumber - required
    - assemblyNumber
    - testerName - required
    - processStep
    - breakOutFullPanel – required
    - userId – required
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- String
    - SUCCESS/ FAIL

130. ProcessTestData

Api/Method:

- *Test/*ProcessTestData

Input Paramters:

- [FromBody]ProcessTestData
  - testData
  - dataFormat
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- `SUCCESS/ FAIL`

Wip

131. AssemblyRouteWipInsert

Api/Method:

- Wip/AssemblyRouteWipInsert

Input Paramters:

- [FromBody]AssemblyRouteWipInsert
  - usrId – required and validated
  - assemId
  - fmaRouteId – FactoryMARoute_ID
  - wipId
  - batchId – BatchID_ID
  - processSingleBoardOnPanel = "0" if process entire panel
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

Note:

132. BoardHistoryReport

Api/Method:

- Wip/BoardHistoryReport

Input Paramters:

- [FromBody]GetBoardLinkedData
  - custId – required and validated
  - serial – serial
  - useMultiPartBarCode -- 0 if no using multiple part barcode
  - lang – 0 if English
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - TestType
  - SerialNumber
  - Number
  - Assembly
  - Revision
  - Version
  - FactoryText
  - MAText
  - RouteText
  - Test_Process
  - TestStatus
  - DefectLocation
  - No_Of_Defects
  - PinCount
  - StartDateTime
  - StopDateTime
  - EquipmentRouteName
  - Material
  - Defect_ID
  - Defect
  - DefectCategory
  - Customer
  - LastName
  - FirstName
  - DefectDetail
  - Memo
  - RepairStatus
  - RepairFlag

- RepairDateTime
- RepairCategory
- RepairedByFirstName
- RepairedByLastName
- DeviationNumber
- DeviationFromDate
- DeviationToDate
- DevitationType
- FixtureText
- FixtureSlot
- FamilyText
- ChildSerialNumber
- ParentSerialNumber
- ODBCTime
- FailureLabel
- MeasuredData
- MeasuredUnit
- DataRec_ID

Preconditions:

- na

Note:

133. GetBoardData

Api/Method:

- Wip/GetWipBySerial

Input Paramters:

- [FromBody]GetWipBySerial
    - custId – custId must be valid and not “0”
    - serial
    - useMultiPartBarCode
    - autoTranslate
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- Wip_ID
- SerialNumber
- Customer_ID

- SeqNumber
- Panel_ID
- Panel
- ProcessAsPanel
- Active
- Purge
- ReferenceUnit
- BirthStatus
- ReferenceUnitActive
- Scrap
- OnHold
- CurrentlyOnHold
- RMA
- RMAReturnCount
- Deviation
- MultiPartBarCode
- BoardPanelBroken
- Assembly_ID
- Number
- Revision
- Version
- BuildStatus_ID
- BuildStatusText
- Deviation_ID
- BatchID_ID
- BatchID
- PreviousAssembly_ID
- NextAssembly_ID

Preconditions:

Note:

134. GetBoardDataAllCustomers

Api/Method:

- Wip/GetWipBySerial

Input Paramters:

- [FromBody]GetWipBySerial
    - serial
    - useMultiPartBarCode

- autoTranslate
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of:
  - Wip_ID
  - SerialNumber
  - Customer_ID
  - SeqNumber
  - Panel_ID
  - Panel
  - ProcessAsPanel
  - Active
  - Purge
  - ReferenceUnit
  - BirthStatus
  - ReferenceUnitActive
  - Scrap
  - OnHold
  - CurrentlyOnHold
  - RMA
  - RMAReturnCount
  - Deviation
  - MultiPartBarCode
  - BoardPanelBroken
  - Assembly_ID
  - Number
  - Revision
  - Version
  - BuildStatus_ID
  - BuildStatusText
  - Deviation_ID
  - BatchID_ID
  - BatchID
  - PreviousAssembly_ID
  - NextAssembly_ID

Preconditions:

Note:

- This differs from GetBoardData in that if the same serial number points to multiple boards, all the boards will return. Serial number is unique by customer only.

135. GetBoardLinkedData

Api/Method:

- Wip/GetBoardLinkedData

Input Paramters:

- [FromBody]GetBoardLinkedData
    - custId – required and validated
    - serial – serial
    - langId – “0” is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
    - Wip_ID
    - ChildWip_ID
    - LinkData_
    - LinkData
    - LinkMaterial_ID
    - LinkObject
    - LinkObjectRevisionUserID_ID
    - LastUpdated
    - IsAssembly

Preconditions:

- na

Note:

- take wip, child wip or unique serial

136. GetParentBoard

Api/Method:

- Wip/GetParentBoard

Input Paramters:

- [FromBody]GetParentBoard
    - custId – required and validated
    - serial – wip serial that is on the panel
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- Wip_ID
- SerialNumber
- SeqNumber
- Panel_ID
- Panel
- Active
- Purge
- BirthStatus
- Scrap
- OnHold
- CurrentlyOnHold
- RMA

Preconditions:

- na

Note:

- take both child wip or unique serial as input

137. GetWipBirthAndCurrentInfo

Api/Method:

- Wip/GetWipBirthAndCurrentInfo

Input Paramters:

- [FromBody]GetWipBirthAndCurrentInfo
    - usrId
    - custId
    - serial
    - birthStatus
    - active
    - langId – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- Wip_ID
- SerialNumber
- Customer_ID

- SeqNumber
- BirthAssembly_ID
- BirthNumber
- BirthRevision
- BirthVersion
- BirthAssembly
- BirthFamily
- BirthDate
- CurrentNumber
- CurrentRevision
- CurrentVersion
- CurrentAssembly
- CurrentFamily
- LastScan
- BirthBatchID_ID
- BirthBatch
- BirthBatchDescr
- CurrentBatchID_ID
- CurrentBatch
- CurrentBatchDescr

Preconditions:

Note:

138. GetWipBatchNextAssembly

Api/Method:

- Wip/GetWipBatchNextAssembly

Input Paramters:

- [FromBody]GetWipBatchNextAssembly
  - custId
  - serial
  - routeStepId
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - NextAssembly_ID

Preconditions:

Note:

139. GetWipBySerial

Api/Method:

- Wip/GetWipBySerial

Input Paramters:

- [FromBody]GetWipBySerial
    - custId – custId must be valid and not "0"
    - serial
    - useMultiPartBarCode
    - autoTranslate
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- Wip_ID
- SerialNumber
- Customer_ID
- SeqNumber
- Panel_ID
- Panel
- ProcessAsPanel
- Active
- Purge
- ReferenceUnit
- BirthStatus
- ReferenceUnitActive
- Scrap
- OnHold
- CurrentlyOnHold
- RMA
- RMAReturnCount
- Deviation
- MultiPartBarCode
- BoardPanelBroken

Preconditions:

Note:

140. GetWipComponentUsageFromComponent

Api/Method:

- Wip/GetWipComponentUsageFromComponent

Input Paramters:

- [FromBody]
  - Material
  - StartTime
  - EndTime
  - Language_ID – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - Assembly
  - Quantity
  - WIP_ID
  - SerialNumber
  - Active
  - Purge
  - Scrap
  - OnHold
  - CurrentlyOnHold
  - RMA
  - Deviation
  - GRN
  - CurrentRoute
  - CurrentStep
  - LastWIPUpdate
  - ParentWIP_ID
  - ParentSerialNumber
  - ParentActive
  - ParentPurge
  - ParentScrap
  - ParentOnHold
  - ParentCurrentlyOnHold
  - ParentRMA
  - ParentDeviation
  - ParentCurrentRoute
  - ParentCurrentStep
  - ParentLastWIPUpdate

Preconditions:

Note:

- Please work with your application admin to make sure sure the web.config of api have the following setting:
    - configuration setting, sqlReportingTimeout, is used to alter SQL command timeout for GetReport method
        - Timeout in seconds.
        - E.g. 300. Please note that 0 will not be accepted. In the case 0 is set, the api will assume default behavior of 30 secs timeout

141. GetWipHoldHistory

Api/Method:

- Wip/GetWipHoldHistory

Input Paramters:

- [FromBody]GetWipHoldHistory
    - custId
    - serial
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
    - CustomerText
    - SerialNumber
    - HoldTypeText
    - HoldTypeDescription
    - HoldDateTime
    - HoldBy
    - HoldMemo
    - ReleaseDateTime
    - ReleaseBy
    - ReleaseMemo

Preconditions:

Note:

- Wip hold history in HoldDateTime descending order

142. ListAssemblyRouteWip

Api/Method:

- Wip/ListAssemblyRouteWip

Input Paramters:

- [FromBody]ListAssemblyRouteWip
  - wipId
  - langId – “0” is not localized or English
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - Customer_ID
  - CustomerText
  - DivisionText
  - Assembly_ID
  - AssemblyText
  - Number
  - Revision
  - Version
  - Family_ID
  - FamilyText
  - FactoryMARoute_ID
  - FactoryText
  - MAText
  - RouteText
  - WIP_ID
  - SerialNumber
  - Active
  - BatchID_ID
  - BatchID
  - UserID_ID
  - LastUpdated
  - CheckDate

Preconditions:

Note:

143. ListCRDs

Api/Method:

- Wip/ListCRDs

Input Paramters:

- [FromBody]
  - wip_Id
  - assembly_Id – optional, can pass 0
  - associateDeviation – optional, can pass blank string
  - partialKey – optional, used to filter CRD list
  - maxCount – optional, can pass 0 to return all results
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List of
  - CRD
  - Material_ID
  - Material
  - PinCount
  - EquipmentOrRouteStep_ID
  - IsEquipment
  - Equipment
  - Feeder
  - Tray
  - Track
  - FeederTrakTrack_ID
  - ComponentType
  - GRN_ID
  - GRN
  - Descr
  - RouteStepSetup
  - RouteStep_ID
  - Equipment_ID

Preconditions:

Note:

- WIP/ListCRDs depends on the MES stored procedure *up_QM_ListCRDsWithMoreInfo*
  Please check if this stored procedure is the May 27 2014 or higher

  (In MES, you can partial key search CRDs in the CRD drop down in manual inspection)
  If this stored procedure is old, you may have to upgrade MES before using this method.

If you use an out of date procedure, you may receive this error:

Error: up_QM_ListCRDsWithMoreInfo has too many arguments specified

144. ListWipMeasuredDataByDataLabel

Api/Method:
- Wip/ListWipMeasuredDataByDataLabel

Input Paramters:
- [FromBody]ListWipMeasuredDataByDataLabel
  - custId
  - serial
  - dataLabel
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:
- List of
  - SerialNumber
  - DataLabel
  - MeasuredData
  - LastUpdated

Preconditions:

Note:
- Wip MeasuredData of specific DataLabel for a wip in LastUpdated descending order.

145. ListWipRouteStepById

Api/Method:
- Wip/ListWipRouteStepById

Input Paramters:
- [FromBody]ListWipRouteStepById
  - wipId
  - langId – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
    - WIP_ID
    - SerialNumber
    - RouteStep_ID
    - FactoryText
    - MAText
    - RouteText
    - StepText
    - Equipment_ID
    - Vendor
    - Model
    - CommonName
    - StartTime
    - EndTime
    - OpStatus_ID
    - OpStatusText
    - StartUserID_ID
    - EndUserID_ID

Preconditions:

Note:

146. ListWipRouteStepBySerial

Api/Method:

- Wip/ListWipRouteStepBySerial

Input Paramters:

- [FromBody]ListWipRouteStepBySerial
    - Serial – wip serial
    - langId – “0” is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- List Of
    - WIP_ID
    - SerialNumber
    - RouteStep_ID
    - FactoryText
    - MAText
    - RouteText

- StepText
- Equipment_ID
- Vendor
- Model
- CommonName
- StartTime
- EndTime
- OpStatus_ID
- OpStatusText
- StartUserID_ID
- EndUserID_ID

Preconditions:

Note:

147. LocateBoard

Api/Method:

- Wip/LocateBoard

Input Paramters:

- [FromBody]LocateBoard
  - custId
  - serial – board, child board or unique linkdata
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- Wip_ID
- SerialNumber
- Customer_ID
- SeqNumber
- Panel_ID
- Panel
- ProcessAsPanel
- Active
- Purge
- ReferenceUnit
- BirthStatus
- ReferenceUnitActive
- Scrap
- OnHold

- CurrentlyOnHold
- RMA
- RMAReturnCount
- Deviation
- MultiPartBarCode
- BoardPanelBroken
- Assembly_ID
- Number
- Revision
- Version
- FamilyText

Preconditions:

Note:

148. PlaceBoardOnHold

Api/Method:

- Wip/PlaceBoardOnHold

Input Paramters:

- [FromBody]PlaceBoardOnHold
  - usrId – required and validated
  - wipId – required
  - holdTypeId – required; used hold type configured for specific operation purpose.
  - processId = "0" if none
  - memo – required; “” if not memo intended
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

- subject to hold validation

Note:

- wip must be valid and still in process. E.g., a packed wip cannot be put on hold.
- Wip cannot be put on hold if it is already on hold for the same hold type.
- Set Process_ID = 0
- Caller client is responsible to figure out HoldType_ID to be used.

149. ReleaseBoardFromHold

Api/Method:

- Wip/ReleaseBoardFromHold

Input Paramters:

- ReleaseBoardFromHold
  - usrID – required and validated
  - wipID – required
  - HoldDateTime – required
  - ReleaseMemo – required; “” if not memo intended
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

Note:

- HoldDateTime and wipId must match a hold record
- If there is no match, an error will occur

150. ScrapWip

Api/Method:

- Wip/ScrapWip

Input Paramters:

- [FromBody]SerialNumberScrap
  - custId
  - serial
  - usrId
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

- Serial number must not have child component

- Serial number must not be packed
- Serial number must no be associated to an order

Note:

151. ValidateWipBarcode

Api/Method:

- Wip/ValidateWipBarcode

Input Paramters:

- [FromBody]ValidateWipBarcode
  - Serial – wip serial
  - Mask – barcode mask
  - custId
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- 0 (fail) or numeric number

Preconditions:

Note:

- Any >0 return is a success.

152. WarehouseWipContainerTransfer

Api/Method:

- Wip/WarehouseWipContainerTransfer+

Input Paramters:

- [FromBody]WipMove
  - custId – required and validated
  - serial – required; board/wip serial
  - sourceContainer – required; container that board/wip is currently stored
  - targetContrainer – required; container that board/wip to be transferred to
  - changeTargetContentRule – if “1”, change the targetContainer’s ContentRule

to 4 (AllBoards); otherwise, set it to “0”
    - routeStepId – optional; if no WipMove for the wip required, set it to “0”
    - equipId – optional; if no WipMove for the wip required, set it to “0”
    - usrId – required and validated
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

- the source container must be in either packed /partial; hold box will be rejected
- target container must be in a state allowing new board (the box has not yet reach its capacity), and its content rule must allow receiving the transferred board. In otherwise, box count < box capacity and content rule will be validated.
- board transfer only occurs if board is allowed in the box per content rule, and if box has not yet reached it capacity .

Note:

- This api is intended for warehouse use only, where boards packed in the originally box are assumed to have been properly process verified according to route and process verification route configuration at the time of original packout.
- The board will be unpacked from the source box; board status will remain unchanged
- After wip transfer, source box status will be changed to partial. Note the original box label is likely no longer applied (box content has changed).
- No process verification or material check is performed when repacking board in the target box.
- Wipmove will occur if a warehouse wip container transfer step is configured, which is reflected by routeStepId and equipId are both supplied.
- Upon transfer, target box will be updated with box count (increased by 1) and its status will be changed to Packed if it has been fulled to its capacity.

153. WipBirth

Api/Method:

- Wip/WipBirth

Input Paramters:

- [FromBody]WipBirth
    - custId – required and validated
    - usrId – required and validated
    - routeStepId
    - equipId

- assemId – required
- serials – wip serials to be birthed, separated by “|”
- panelId = “0” or CR_Panels.Panel_ID
- panelSerial = “” – wip serial used as panel identifier
- batchId = "0"if none
- deviationId = "0" if none

- sqlServer = "" – optional
- dataBase = "" – optional

<u>*Expected Result*</u>:

- success/error

<u>*Preconditions:*</u>

- Panel has been configured in MES
- Route and assembly-route association configured in MES

Note:

- Validation includes assembly, panel, serial barcode, etc
- Birth multiple WIPs at once
- If panelId <> 0, serials enter must not exceed panel capacity
- Inserts data in following tables
  - dbo.WP_BatchAssemblies
  - dbo.WP_Wip
  - dbo.WP_Panels
  - dbo.WP_AssemblyRouteWip
  - dbo.WP_WipRouteSteps
  - dbo.QM_DeviationWip

154. WipMove

<u>*Api/Method:*</u>

- Wip/WipMove

<u>*Input Paramters*</u>:

- [FromBody]WipMove
  - usrId – required and validated
  - wipId – required
  - routeStepId – required
  - equipId – required
  - withPV – whether PV to be performed
  - opStatusId = "1"
  - ioType = "0" if in and out
  - newAssemId = "0” if not for assembly progression

- procSingleBoardOnPanel = "1"; whether to process single board; should use default
    - commonName = "" =testerName; if withPV=1, commonName must be valid
    - langId – "0" is not localized or English – optional
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:
- success/error

Preconditions:
- wip must have been created/birthed
- route/routestep/pv configured done and validated in MES

Note:
- This function would need be called before linking. Please see LinkWithValidation for more details.
- opStatus: 1=Pass, 2=Fail; 3=Abort; 0=Present.
- ioType: 0=In/Out; 1=In; 2=Out
- return PV error: ["RuleText"] + " (" + ["StepText"] + " / " + ["DescrText"] + " -- " + ["ParameterText"] + ")"

155. WipMoveBySerial

Api/Method:
- Wip/WipMoveBySerial

Input Paramters:
- [FromBody]WipMoveBySerial
    - usrId – required and validated
    - customer – required
    - division – required
    - serial – wip serial
    - factory – required
    - ma – required
    - route – required
    - stepDescr – required
    - equipment – =testerName; if withPV=1, commonName must be valid
    - withPV – whether PV to be performed
    - opStatusId = "1"
    - ioType = "0"
    - breakoutPanel

- newAssemId = "0” if not for assembly progression
- procSingleBoardOnPanel = "1”; whether to process single board; should use default
- langId – “0” is not localized or English – optional

- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

- wip must have been created/birthed
- route/routestep/pv configured done and validated in MES

Note:

- opStatus: 1=Pass, 2=Fail; 3=Abort; 0=Present.
- ioType: 0=In/Out; 1=In; 2=Out
- return PV error: ["RuleText"] + " (" + ["StepText"] + " / " + ["DescrText"] + " -- " + ["ParameterText"] + ")"

156. WipRouteStepInsert

Api/Method:

- Wip/WipRouteStepInsert

Input Paramters:

- [FromBody]WipRouteStepInsert
  - wipId
  - routeStepId
  - equipId
  - ioType
  - opStatusId
  - panelType -- 0 to 3 numeric value
  - processAsPanel
  - wipMoveOnly
  - startUsrId = "0”
  - endUsrId = "0”
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

- success/error

Preconditions:

Note:

157. Wipscandata

Api/Method:

- Wip/WipScanData

Input Paramters:

- [FromBody] `ListWipScanData`
    - RouteStep
    - StepInstance
    - StartDateTime
    - EndDateTime
    - LangId
- sqlServer = "" – optional
- dataBase = "" – optional

Expected Result:

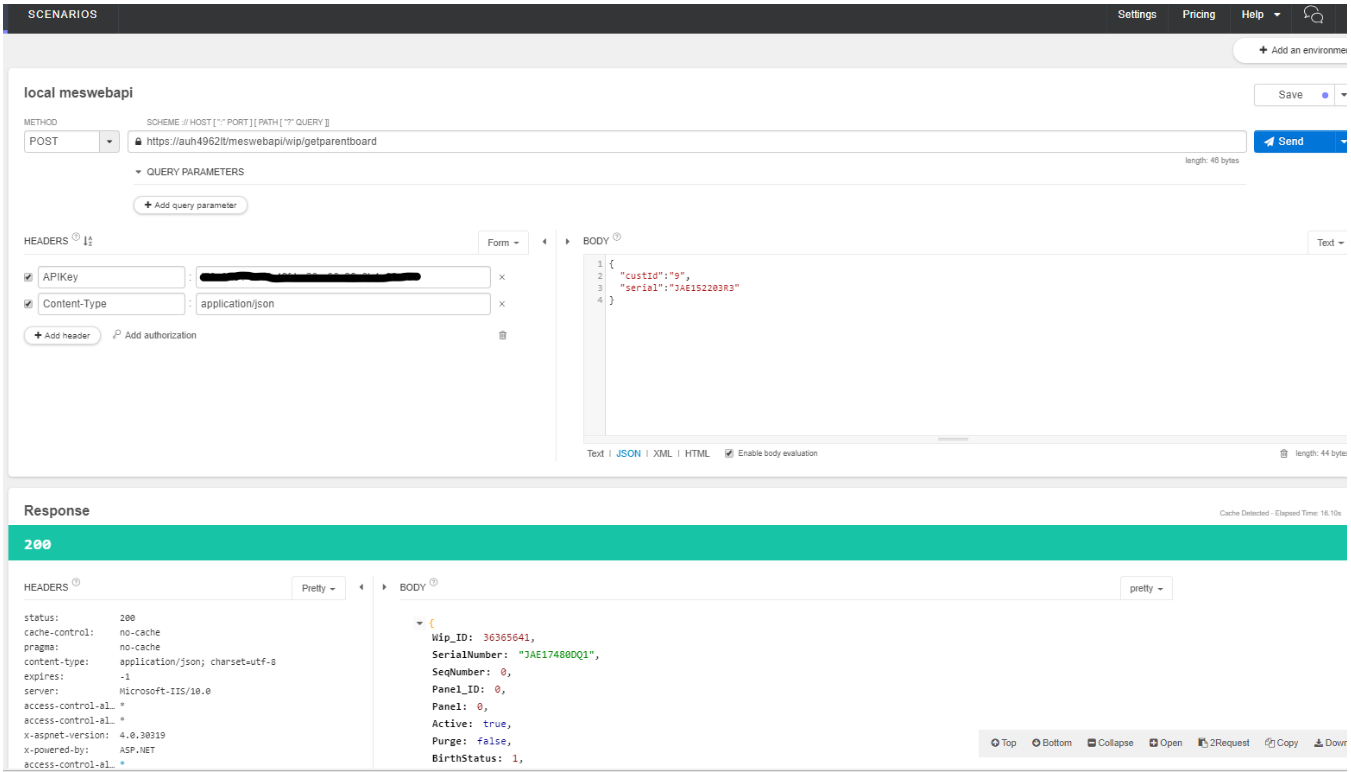
- Site
- Building
- ManufacturingArea
- ManufacturingAreaId
- Route
- RouteId
- Step
- StepId
- Equipment
- EquipmentId
- Customer
- Division
- Assembly
- Revision
- ProcessLoop
- TestLoop
- TestStatus
- ComlpetionTime

• WipId

Test with Rest Client

The following example is using Restlet Client/Chrome

url: https://AZAPSEPENTRM41



Example of Api response

[
{
"Wip_ID": 1144749,
"ChildWip_ID": 0,
"LinkData_ID": -4398046506876,
"LinkData": "BLANK 001",
"LinkMaterial_ID": -1999999753,
"LinkObject": "N20-B6620-1-S4",
"LinkObjectRevision": "",
"UserID_ID": 221,
"LastUpdated": "2009-08-14 10:36:39.950",
"IsAssembly": 0
},
{
"Wip_ID": 1144749,
"ChildWip_ID": 0,
"LinkData_ID": -4398046506875,

"LinkData": "BLANK 002",
"LinkMaterial_ID": -1999999752,
"LinkObject": "N20-B6620-1-S5",
"LinkObjectRevision": "",
"UserID_ID": 221,
"LastUpdated": "2009-08-14 10:36:40.060",
"IsAssembly": 0
},
{
"Wip_ID": 1144749,
"ChildWip_ID": 0,
"LinkData_ID": -4398046506874,
"LinkData": "BLANK 003",
"LinkMaterial_ID": -1999999751,
"LinkObject": "N20-B6620-1-S6",
"LinkObjectRevision": "",
"UserID_ID": 221,
"LastUpdated": "2009-08-14 10:36:40.153",
"IsAssembly": 0
},
{
"Wip_ID": 1144749,
"ChildWip_ID": 0,
"LinkData_ID": -4398046506873,
"LinkData": "BLANK 004",
"LinkMaterial_ID": -1999999750,
"LinkObject": "N20-B6620-1-S7",
"LinkObjectRevision": "",
"UserID_ID": 221,
"LastUpdated": "2009-08-14 10:36:40.250",
"IsAssembly": 0
}
]

System/Software Specification

Application/Service

- Name: MESWebApi
  - New website/api
  - Published in IIS (OS targeted: Windows 2012 R2)
  - Read/write data to MES JEMS database via api methods listed in this document
- Use rules:
  - For approved client/callers only
    - Approved callers will be issued with APIKey for connection purpose (see web security section) for the details
    - The site IT/TE/other developers are forbidden to share APIKey without written approval
- Additional Note:
  - Please following Jabil Security/Data rule to secure production data
  - Please escalate if we have concern on the use of MESWebApi.

Installation/Publishing

- Target: IIS (OS targeted: Windows 2012 R2)
- Website/service publish – Https + certificate:
  - Https:
    - bidirectional encryption between client and server – prevent eavesdropping and tempering
  - Server certificate
    - Encrypted by Secure Sockets Layer (SSL)
    - Certificate is used to initiate "SSL handshake", which generates shared secrets to establish a uniquely secure connection between caller and the website.

Web Sucurity/Software Solution

- Message Handler:
  - We issue APIKey to approved callers
  - implement delegating "MessageHandler" to validate caller and address cross cutting concerns
  - Client application are required to include "APIKey" and value in all its api calls to MESWebApi.
- Validate MES user
  - All write functions to MES will require valid MES user.
  - This is done by

        - Request a domain service account for Quake shop floor equipment to share
        - Configure this service account as MES user
        - Use this service account's MES identitity for all the "write" operation by passing in valid userid_id.
- Security Goal:
    - Https + Certificate + APIKey(in request headers) + UserId (in all write api calls) will give us sufficient security without hindering performance.

--End of document--